

The Importance of Being Slow: The Costs and Benefits of Phasing-In Regulatory Reforms

Gianni De Nicolò* Nataliya Klimenko[†] Sebastian Pfeil[‡]
Jean-Charles Rochet[§]

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*The Johns Hopkins Carey Business School and CESifo, 100 International Drive, Baltimore, MD 21202-1099, USA, Phone: +1 4102344507, E-mail: gdenico1@jhu.edu.

[†]University of Zürich, Plattenstr. 32, 8032 Zürich, Switzerland, Phone: +41 446344055, E-mail: nataliya.klimenko@bf.uzh.ch.

[‡]University of Groningen, P.O. Box 800, 9700 AV Groningen, The Netherlands, Phone: +31 6 50 01 98 67, E-mail: c.s.pfeil@rug.nl (corresponding author).

[§]Toulouse School of Economics, 1 Esplanade de l'Université, 31080 Toulouse Cedex 06, France, Phone: +33 5 61 12 86 18, E-mail: jean-charles.rochet@tse-fr.eu.

Abstract

We build a stylized dynamic equilibrium model with financial frictions to study how capital requirements affect the dynamics of aggregate bank lending. A sudden regulatory tightening may cause substantial issuance of bank capital, but also a severe credit crunch. When regulatory tightening is introduced with a transition phase, the anticipation of larger profits from lending encourages banks to retain more earnings during the transition phase. This accelerates the accumulation of capital buffers and reduces the risk of a credit crunch. In an intermediate range of aggregate bank capital, the same anticipation effect can cause loan rates to decrease and lending to increase in response to the announcement of a regulatory tightening, consistent with the rise in aggregate lending documented after the announcement of Basel III. We characterize the welfare-maximizing length of the transition phase. Without the anticipation effect, a delay only shifts the switch between the two regimes in time. Anticipation makes the transition phase itself valuable and accounts for up to almost half of the welfare gain from tighter regulation: a regulator who does not account for it chooses a transition that is too short and forfeits up to two fifths of the total welfare gain from the tightening.

Keywords: Bank capital requirements, credit crunch, systemic risk

JEL Classification: E21, E32, F44, G21, G28

1 Introduction

The 2007–2009 financial crisis has provoked a remarkable sequence of events. Initially, large losses eroded the capital buffers of banking systems around the world. Many banks were recapitalized by a combination of government interventions (e.g. Laeven & Valencia, 2013), and capital injections by shareholders (e.g. Black et al., 2016; Homar & van Wijnbergen, 2017), which was followed by severe credit crunches and reductions in real economic activity (the “Great Recession”) (e.g. Aiyar et al., 2016; De Jonghe et al., 2020; Fraise et al., 2020). The Basel Committee has in turn introduced new regulatory guidelines (Basel III), which have been implemented over a transition phase of several years.¹

Our paper studies the costs and benefits of phasing-in a regulatory tightening over a transition phase.² The common rationale for phasing-in a regulatory tightening is to allow banks to gradually build up capital buffers, and thus avoid the reduction in lending following an immediate implementation. We find that this strategy is effective because banks adjust their issuance and payout policies during the transition phase: Anticipating that capital will be more valuable after the implementation of the regulatory tightening, banks retain more of their earnings and convert them to book equity. This helps to achieve the financial stability gains from tighter regulation more quickly and to reduce the risk of a credit crunch. As the anticipation effect thus increases the value of being slow, regulators who did not properly take into account the anticipation-induced change in bank policies would significantly underestimate the welfare maximizing length of the transition phase. Furthermore, the anticipation of lending being more restricted after the implementation of a regulatory tightening affects the risk premium required by banks. Quite counter-intuitively, in an intermediate range of aggregate bank capital, this may cause lending to increase during the transition phase. This prediction is consistent with the finding that aggregate lending increased after the announcement of Basel III (see Basel Committee on Banking Supervision, 2022, Section

¹As documented in World Bank (2020), minimum capital requirements have steadily increased worldwide since the 2007–2008 financial crisis. Basel III, announced in 2010, phased in the higher capital requirements between 2013 and 2019 (Basel Committee on Banking Supervision, 2011, Annex 4); its final standards, agreed in 2017, were again announced with a transition period, taking effect in 2022 and phased in over a further five years (Basel Committee on Banking Supervision, 2017).

²Most of the literature has studied the optimal level of capital requirements, but not its optimal timing (cf. e.g. Admati et al., 2013; Basel Committee on Banking Supervision, 2019; DeAngelo & Stulz, 2015; Gorton & Winton, 2017).

8.1.3 and Annex A.18); Section 5 relates the model’s cross-sectional predictions to US bank data.

In order to analyze how the introduction of (tighter) capital requirements affects aggregate lending, interest rates, and capital buffers, we endogenize the dynamics of bank capital in a stylized equilibrium model of banking. Banks finance risky loans to the real sector by issuing equity and liquid deposits, which households use for transactions.³ The two main frictions in the model are: (i) households cannot invest directly in the real sector and (ii) banks incur a flotation cost when they issue equity. This implies that banks will retain earnings to cover future losses on their loan portfolios and save on refinancing costs. Banks are subject to a simple capital requirement, according to which a minimum fraction of loans must be financed by equity. The motivation for capital requirements in our model is the risk of systemic crises with socially costly bank failures.

The equilibrium loan rate decreases in aggregate bank capital. This reflects the fact that the aggregate supply of bank loans increases with aggregate bank capital and, as is standard, aggregate loan demand decreases in the loan rate. Lending, therefore, becomes less profitable when banks build capital buffers by converting retained earnings into book equity. Since loans are partially financed by equity, the market-to-book ratio of equity capital must therefore also decrease in aggregate bank capital. This implies that even though the loan market is perfectly competitive, banks earn a strictly positive lending premium, akin to the risk-premium required by a risk-averse individual with decreasing marginal utility.

Bank strategies in our model resemble the optimal corporate policies in partial equilibrium models such as Bolton et al. (2011); Bolton et al. (2013), Décamps et al. (2011), or De Nicolò et al. (2014), where issuance costs in combination with (exogenous) costs to hold liquidity lead to a decreasing marginal value of cash (i.e. firms become effectively risk averse). However, in our model, the costs and benefits of accumulating equity are determined endogenously in a market equilibrium. When the level of aggregate capital and thus loan supply is sufficiently high, the equilibrium loan rate is so low that the marginal loan is unprofitable. Hence, instead of further expanding their loan making businesses, banks optimally distribute any further earnings to their shareholders at that point. If, on the other

³In contrast to existing models such as Brunnermeier and Sannikov (2014), in which financial firms themselves manage productive assets in the economy, we explicitly take into account the intermediation activity by banks.

hand, the aggregate capital is sufficiently low, the aggregate loan supply is restricted by the binding capital requirement. This implies a high equilibrium loan rate, which makes lending to the real sector very profitable. Despite having to bear flotation costs, it is then optimal for banks to issue new equity claims to avoid further deleveraging. Hence, bank policies follow a barrier strategy. At the lower, or issuance boundary, the market-to-book ratio reaches its maximum equal to the total marginal issuance costs. At the upper, or payout boundary, the market-to-book ratio reaches its minimum equal to shareholders' marginal utility of consuming the paid out earnings. In between the two barriers, all earnings are retained to build up capital buffers which, in turn, are used to cover losses on the asset side. Our theory of bank capital therefore emphasizes its loss-absorbing role instead of its incentive effects.⁴

With tighter capital requirements, equity capital becomes more valuable for banks (reflected by a higher market-to-book ratio) for two reasons. First, more capital is needed to finance a given level of lending. Second, lending is more profitable as the aggregate supply of bank loans is more severely restricted.⁵ With equity capital thus being more valuable under tighter capital requirements, banks optimally retain more earnings and are more inclined to issue new claims despite the issuance costs. That is, both the payout boundary and the issuance boundary are higher under tighter regulation. A regulatory tightening without a transition phase may therefore trigger an immediate recapitalization up to the new issuance boundary. However, it can also cause a severe credit crunch when the tighter regulation becomes binding upon implementation.

If a regulatory tightening is announced and phased in over a transition phase, banks know that it will take effect and adjust their behavior before it does. The anticipation of loan supply being more restricted under tighter regulation is “priced-in” by banks if the banking sector is very well capitalized. This leads to a higher equilibrium loan rate and thus a lower level of aggregate lending. However, if the banking sector is poorly capitalized, aggregate lending is fully determined by the binding capital requirement and therefore unaffected by the announcement of a regulatory tightening. Hence, if aggregate capital increases, banks move closer toward the region where lending is more profitable since the anticipated scarcity of loans is priced-in for high levels of aggregate capital. This implies that in the intermediate

⁴This appears to capture regulators' motives to impose capital requirements which restrict banks' total equity (i.e. their capital structure). What matters for bank managers' risk-taking incentives, by contrast, is a bank's inside equity including compensation packages.

⁵A related point is made in Schliephake (2016).

region, where the regulatory requirement becomes slack, the market-to-book ratio of equity decreases more slowly in aggregate capital. That is, just as the utility function of a risk-averse individual becomes less concave, the lending premium required by banks decreases in response to the regulatory announcement in this region. As a result, lending will increase, even compared to the status quo without a regulatory reform. Similarly, banks also re-balance their loan portfolios towards more risky loans. We relate these predictions to the lending and loan rates of US bank holding companies before and after the announcement of Basel III: lending rose and loan rates fell during the transition, and among banks with intermediate capitalization, lending grew more at those operating in riskier markets.

Moreover, as banks anticipate capital to be more valuable after the implementation of tighter regulation, they want to accumulate more capital during the transition phase. More precisely, since the anticipated increase in the value of capital is reflected in its current value only for high levels of capital, the additional capital is generated by retaining more earnings rather than issuing more claims.

Finally, we study the welfare-maximizing length of the transition phase. The cost of a slower implementation is that with a longer transition phase, it takes longer for the financial stability gains in terms of larger capital buffers to materialize, and that the exposure of the public backstop during the transition depends on how banks respond to the announcement. The benefit is that banks have more time to accumulate capital so that tighter capital requirements are less likely to be binding when they are implemented. The fact that banks change their policies in response to the regulatory announcement tilts this trade-off towards a longer transition phase: Capital accumulation is accelerated as banks retain more earnings during the transition phase, which tends to reduce the costs of a slower introduction and increase the benefits. The anticipation effect discussed in this paper thus increases the value of being slow. We show that regulators who did not properly take into account how banks adjust their policies when expectations about regulatory reforms change would severely underestimate the optimal transition phase and forfeit up to two fifths of the welfare gain from the tightening. The reason is that, for such a regulator, delaying the reform only shifts the switch between the two regimes in time. Until the switch, the distribution of bank capital merely approaches its stationary distribution under the old regime, so that in our baseline calibration this regulator implements at once when banks are poorly or well capitalized, and chooses a short transition at intermediate capitalization. The anticipation

effect makes the transition phase itself valuable, through the capital that banks build ahead of implementation, and lengthens the optimal transition at every level of capitalization. In our baseline calibration, it accounts for up to almost half of the welfare gain from the tightening. The value of an optimally chosen transition is largest at intermediate levels of bank capital, where an immediate implementation would forfeit most of the welfare gain from the tightening, and not where banks are least capitalized. Furthermore, the value of being slow is lower (the optimal transition phase is shorter) if the social costs of bank failures are higher. Intuitively, with bank failures being more costly, the regulator becomes relatively more concerned about a faster realization of financial stability gains than the stabilization of aggregate lending. If banks have already accumulated larger equity buffers at the time of the regulatory announcement, on the contrary, regulators are less concerned about the quick realization of gains in financial stability. At the same time, however, they are also less concerned about the stabilization of lending, as the tighter capital requirement is less likely to become binding even if the regulatory change is introduced quickly. In our baseline calibration, the optimal transition takes about three years when banks are poorly or moderately capitalized at the time of the announcement, and it shortens to about one and a half years as initial capital approaches the level at which banks pay dividends. The second effect thus dominates only when banks are well capitalized.

Related Literature. First, our paper is related to the academic literature that views capital requirements as a way to trade off the expected social cost of bank failures (which is not internalized by bankers) and the welfare reduction due to the limitations on banks' deposits and lending activities. Admati et al. (2013) argue that the first effect dominates and that capital requirements should be much higher. In contrast, DeAngelo and Stulz (2015); Diamond and Rajan (2000) emphasize the second effect and argue that high capital requirements impede banks' provision of liquidity. Van den Heuvel (2008) was the first to develop a dynamic general equilibrium model that allowed a quantitative evaluation of this trade-off. By calibrating his model on US data, he finds that the social cost of capital requirements amounts to a permanent reduction of aggregate consumption of 0.1 to 1 percent. Martinez-Miera and Suarez (2014) also analyze the impact of capital requirements on bank

risk taking incentives in a quantitative dynamic general equilibrium model.⁶

Our model, however, emphasizes the loss absorbing role of bank capital, instead of its incentive effects. In practice, banks typically maintain equity ratios well in excess of regulatory capital requirements,⁷ which can only be understood as precautionary buffers against a future need for a costly issuance of new capital. Milne and Whalley (2001) were the first to explore a simple dynamic model with this feature. They show that in the long run, capital requirements have no impact on banks' risk taking. Allen et al. (2011) elaborate on the role of capital buffers by showing how they allow banks to commit to monitoring loans, which ultimately benefits borrowers. Finally, also Corbae and D'Erasmus (2021) consider a model that emphasizes the loss absorbing role of bank capital and analyze the relation between regulation and industry structure.

Following He and Krishnamurthy (2012) and Brunnermeier and Sannikov (2014), a new strand of the literature has developed dynamic macroeconomic models with financial frictions and shown that the (endogenous) capitalization of the financial sector is a crucial factor in explaining the performance of the economy. Closely related work develops dynamic models in which intermediary balance-sheet constraints and leverage cycles drive procyclical credit supply and time-varying intermediary risk premia (Adrian & Boyarchenko, 2015; Gertler & Karadi, 2011). In Kondor and Vayanos (2019), arbitrageurs offer risk management services to hedgers. In equilibrium, the wealth of arbitrageurs is a priced risk factor that influences market risk aversion. In Phelan (2016), banks cannot issue new equity but must rely on retained earnings to increase their capital buffer. As a result, aggregate outcomes depend on the endogenously determined total equity of the banks. In Bolton et al. (2025), banks cannot perfectly control their deposit flows. Precautionary equity buffers are needed to limit the risk that sudden deposits inflows can force a bank to issue costly equity. Banks' risk aversion is also endogenously determined. Our model exhibits similar features: banks' risk aversion and loan rates are endogenous functions of the total capitalization of the banking sector. The stationary equilibrium of our model builds on the aggregate-bank-capital framework of Klimenko et al. (2017), in which a single loan market clears and there is no capital regulation; what is new here are the two loan segments, the non-stationary equilibrium

⁶The incentive role of (inside) equity is studied in a large literature including for instance Hellmann et al. (2000); Morrison and White (2005); Repullo (2004).

⁷Fonseca and González (2010) show how these capital buffers vary across countries.

during the transition to an announced requirement, and the welfare-optimal length of that transition.

Our paper is also related to the recent literature that has developed dynamic general equilibrium models to examine the long-term impact of capital requirements, such as Begenau and Landvoigt (2022); Clerc et al. (2015); Davydiuk (2017); Elenev et al. (2021); Sundaresan and Wang (2023). Closest to our question, Mendicino et al. (2020) show that the transitional costs of raising capital requirements can exceed their long-run costs and depend on how the increase is phased in; in their analysis the phase-in schedule is a given, whereas we make the delay between announcement and implementation the object of choice and let banks' payout and issuance policies respond to the announcement. Begenau (2020) analyzes the long-term impact of capital requirements on bank lending in a quantitative general equilibrium model. She finds that imposing tighter capital requirements tends to reduce banks' demand for deposits, which drives down interest rates paid on deposits, and thus banks' funding costs. As a result, bank lending can increase in the long run.⁸ In our paper, the long-term impact on lending is negative, albeit relatively small. Instead, we stress that the anticipation of a future regulatory tightening lowers the equilibrium risk-premium and hence increases lending in the *short-run*. Our paper is also related to the recent literature that analyzes the impact of new banking regulations such as liquidity requirements (Hugonnier & Morellec, 2017), the procyclical effects of static capital regulation (Repullo & Suarez, 2013), or countercyclical capital requirements (Malherbe, 2020).

Our paper is finally related to the empirical literature on how banks adjust to higher capital requirements. Gropp et al. (2019) and Mésonnier and Monks (2015) find that banks in the 2011–2012 EBA capital exercise raised their capital ratios by shrinking assets rather than by issuing equity, and Behn et al. (2016) report a comparable contraction in lending under model-based capital regulation in Germany. Our model produces the same adjustment where the requirement binds: since equity issuance is costly, banks adjust on the asset side, and lending is determined by capital. The banks in those episodes are the ones short of capital relative to the target. The lending expansion we emphasize arises instead in the intermediate region, where the current requirement is still slack and only the anticipated one binds. In this range, the announcement also makes banks effectively less risk-averse, so that

⁸In Begenau and Landvoigt (2022), a similar mechanism leads to an increase in risk taking by shadow banks when commercial banks face tighter capital requirements.

lending expands most in the high-risk market (Section 5); consistent with this, Section 5.2 finds that during the Basel III transition, lending grew more at moderately capitalized banks operating in riskier markets.

The remainder of the paper is organized as follows. Section 2 introduces the model, and Section 3 solves the competitive equilibrium. Section 4 discusses the properties of this equilibrium for a given level of capital regulation. Section 5 studies the non-stationary equilibrium in the transition phase and provides empirical evidence on the introduction of Basel III. Section 6 derives the optimal length of the transition phase, and Section 7 concludes. The omitted proofs are given in Appendix A, Appendix B provides additional material, and Appendix C describes the numerical implementation.

2 Model

Households, Banks, and Firms. We consider a stylized dynamic equilibrium model that captures, in the simplest possible way, the role of banks in the economy. Banks provide payment services to households and extend loans to the real sector. Time is continuous with infinite horizon, and there is a single physical good that can be consumed or invested. There is a continuum of risk-neutral households, a continuum of banks and a continuum for each of two different types of short-lived, risk-neutral entrepreneurs, all of mass one.

Banks are financed by equity e_t and deposits d_t . If banks issue new equity to households, they incur a proportional flotation cost of γ , e.g., from brokerage commissions or underwriting fees, as is common, for instance, in the literature on corporate liquidity management (Bolton et al., 2011; Décamps et al., 2011), but also in dynamic models of banking, such as Hugonnier and Morellec (2017). Deposits are fully insured, so the deposit rate equals the common discount rate, $r = \rho$. Deposits are determined residually from the balance-sheet identity (3), $d_t = k_t^l + k_t^h - e_t$.

Dynamics of Bank Capital. Banks can invest in risky assets k_t^l and k_t^h , which represent loans to low-risk and high-risk entrepreneurs, respectively. Loan rates are given by R_t^l and R_t^h , such that the instantaneous return on loans of type $i \in \{l, h\}$ equals:

$$R_t^i dt - \sigma^i dZ_t^i - \phi dN_t. \quad (1)$$

Asset risk contains a continuous component represented by Brownian motions Z^l and Z^h ,⁹ and a jump component represented by a Poisson process N with intensity ζ .¹⁰ The two loan segments are exposed to independent Brownian shocks, so that diversifying across segments reduces the volatility of bank capital, whereas the systemic jump N is common to both. The latter captures the risk of systemic banking crises, which occur at the jump times of N , denoted by $(t_k)_{k \geq 1}$. A systemic crisis destroys the fraction $\phi < 1$ of bank assets, sufficiently large to wipe out all its equity.¹¹ With an equity value of zero, the bank enters a resolution process, which will lead to a sale to new shareholders once the systemic crisis is over. The deposit insurance agency covers the shortfall on deposits. The social costs in case of a bank failure are given by

$$s_t := \theta(\phi(k_t^l + k_t^h) - e_t), \quad (2)$$

where $\theta \geq 0$ is a parameter that measures the relation between the size of the required bailout and the associated social costs of a crisis.¹² A bank's investment in corporate loans k_t^i is financed by equity capital e_t and deposits d_t , which is reflected by the accounting identity:

$$k_t^l + k_t^h = d_t + e_t. \quad (3)$$

The book value of bank capital thus evolves according to

$$\begin{aligned} de_t = & -\rho d_t dt + di_t - dc_t \\ & + k_t^l(R_t^l dt - \sigma^l dZ_t^l - \phi dN_t) \\ & + k_t^h(R_t^h dt - \sigma^h dZ_t^h - \phi dN_t), \end{aligned} \quad (4)$$

⁹ Capturing e.g. changes in total factor productivity (see Brunnermeier & Sannikov, 2014).

¹⁰ We treat ζ as constant, so that capital lowers the cost of a systemic crisis but not its probability. This matches the long-run evidence: across 17 advanced economies since 1870, higher bank capital ratios do not reduce the incidence of financial crises but shorten the recessions that follow them (Jordà et al., 2021).

¹¹ The complementary case, in which a crisis is only sometimes fatal so that a bank may survive it, gives rise to a more complex problem for bank behavior and is the focus of Hoffmann and Pfeil (2021). Section 4 verifies that the calibrated value of ϕ satisfies this restriction.

¹² Imposing a common loss fraction ϕ across loan types is a simplification without consequence for bank behavior: because a systemic crisis wipes out bank equity regardless of portfolio composition, ϕ enters only the social cost (2), not the banks' problem. The two loan types differ in risk through their loadings $\sigma^l < \sigma^h$ in (1).

where the first line reflects the fact that book equity increases with newly issued shares (di_t), while it is reduced by interest payments on deposits (ρd_t) and by payouts to the bank's shareholders (dc_t).¹³ The second and third lines capture the loss-absorbing role of bank equity: Bank capital grows if the net return on corporate loans is positive and decreases if the return is negative. Substituting the residual deposits $d_t = k_t^l + k_t^h - e_t$ from (3), equation (4) can be rewritten as

$$\begin{aligned} de_t &= e_t \rho dt + di_t - dc_t \\ &+ k_t^l [(R_t^l - \rho)dt - \sigma^l dZ_t^l - \phi dN_t] \\ &+ k_t^h [(R_t^h - \rho)dt - \sigma^h dZ_t^h - \phi dN_t]. \end{aligned} \quad (5)$$

Summing up over all banks in the economy, we obtain the dynamics of aggregate bank capital E_t

$$\begin{aligned} dE_t &= E_t \rho dt + dI_t - dC_t \\ &+ K_t^l [(R_t^l - \rho)dt - \sigma^l dZ_t^l - \phi dN_t] \\ &+ K_t^h [(R_t^h - \rho)dt - \sigma^h dZ_t^h - \phi dN_t], \end{aligned} \quad (6)$$

where K_t , dI_t and dC_t stand, respectively, for the aggregate volumes of lending, equity issuance and payments to shareholders at time t . Accordingly, the aggregate social cost of a systemic crisis at time t is

$$S_t := \theta [\phi (K_t^l + K_t^h) - E_t], \quad (7)$$

the shortfall that the deposit insurance agency must cover, scaled by θ . Banks are subject to a regulatory capital requirement Λ , which specifies banks' maximum admissible asset-to-equity leverage, i.e.,

$$a_t^l + a_t^h \leq \Lambda, \quad (8)$$

where leverage (asset-to-equity ratio) for loans of type $i \in \{l, h\}$ is given by

$$a_t^i := \frac{k_t^i}{e_t}.$$

¹³Payouts may take the form of a special dividend or a share repurchase.

An alternative interpretation of (8) is that at least a fraction $1/\Lambda$ of a bank's total assets must be financed by equity. Since banks' assets consist only of loans to firms, which carry a uniform risk weight of 100% under the standardized approach of the Basel framework when the borrower is unrated (Basel Committee on Banking Supervision, 2006, para. 66), (8) is at the same time a requirement in terms of risk-weighted assets. The motivation for capital requirements is to limit the social costs of bank failures in case of a systemic banking crisis given by (7).¹⁴

Demand for Bank Loans. Entrepreneurs finance production of the consumption good by bank loans. Due to technological or informational frictions, households cannot invest directly in the productive sector, but only indirectly via bank equity or deposits.¹⁵ The productive sector consists of *high-risk* and *low-risk* entrepreneurs with demand functions

$$L^i(R^i) = \alpha^i - \beta^i R^i, \quad (9)$$

for $i \in \{h, l\}$, with $\beta^h \leq \beta^l$, such that lending to high-risk entrepreneurs may be more profitable for the bank. As is standard, the total demand for each type of bank loans is thus strictly decreasing in the respective loan rate. Furthermore, we normalize lending in the frictionless benchmark case, where the loan rate equals the risk-free rate ρ , i.e., $L^h(\rho) = L^l(\rho) = 1$.

Having two loan types that differ in risk allows us to model how banks choose the *composition* of their loan portfolio, and not only its *size*, so that the model speaks to bank risk-taking as well as to the volume of aggregate lending. This margin delivers one of our more counterintuitive results: in an intermediate range of aggregate capitalization, the anticipation of a tighter capital requirement makes banks effectively *less* risk-averse and leads them to rebalance *towards* high-risk loans (Section 5), contrary to the common intuition that increasing banks' regulatory "skin in the game" should reduce their appetite for risk. Because this portfolio rebalancing is observable in bank-level data, it also yields a cross-sectional implication of the anticipation channel, which we confront with the data in Section 5.2.

¹⁴See e.g. Miles et al. (2013).

¹⁵See e.g. Freixas and Rochet (2023), Chapter 2.

3 Competitive Equilibrium

Equilibrium Conditions. A competitive equilibrium is defined by a map from shock histories $\{Z_s, N_s, s \in [0, t]\}$, to the loan rates and bank capital such that individual banks maximize their shareholder value and the loan market clears. The dynamic strategy of each individual bank consists of a lending policy, dividend distributions and equity issuance as a function of the bank's individual capital e_t and the aggregate capital E_t . We focus on Markov equilibria, in which the equilibrium loan rates, R_t^l and R_t^h , are deterministic functions of aggregate bank capital. Furthermore, we assume that banks' shareholders form rational expectations and, in particular, correctly anticipate that loan rates are indeed deterministic functions of banks' aggregate equity.

A given bank's shareholder value maximization problem is given by

$$\begin{aligned} v(e_t, E_t) = \max_{k_t^l, dc_t, di_t} \mathbb{E} \left[\int_t^\tau e^{-\rho(s-t)} (dc_s - (1 + \gamma) di_s) \right] \\ \text{s.t. } k_t^l + k_t^h \leq \Lambda e_t. \end{aligned} \quad (10)$$

While aggregate bank policies K_t^l , K_t^h , dC_t , and dI_t , are determined as the sums of banks' individual policies k_t^l , k_t^h , dc_t , and di_t , banks are competitive and take all aggregate variables as given. Each bank is run until the stochastic default time $\tau := \inf\{t : e_t \leq 0\}$, which denotes the first time when the book value of its equity falls to or below zero. At the equilibrium of our model, banks only default if there is a systemic crisis. When flotation costs γ are not too high (which we will assume), shareholders find it indeed optimal to inject fresh equity before book equity falls to zero, such that

$$e_t > 0, \quad (11)$$

for all $t \in \{\mathbb{R}^+ \setminus (t_k)_{k \geq 1}\}$.

Definition 1. A competitive Markov equilibrium is such that

- (i) the loan rates are deterministic functions of aggregate bank capital: $R_t^i = R^i(E_t), i \in \{l, h\}$;
- (ii) banks maximize their shareholder value (10), taking all aggregate variables as given;

(iii) individual and aggregate bank capital follow (5) and (6) with initial conditions e_0 and E_0 ;

(iv) the market for bank loans clears: $K_t^i = L^i(R_t^i), i \in \{l, h\}$.

From Itô's Lemma, the change of variables formula for jump processes, and the dynamics of capital in (5) and (6), it follows that banks' shareholder values have to satisfy the following Hamilton-Jacobi-Bellman equation:

$$\begin{aligned}
\rho v = \max_{k^i, dc, di} & \left\{ [e\rho + (R^l - \rho)k^l + (R^h - \rho)k^h + di - dc]v_e \right. \\
& + [E\rho + (R^l - \rho)K^l + (R^h - \rho)K^h + dI - dC]v_E \\
& + ((k^l)^2v_{ee} + (K^l)^2v_{EE} + 2k^lK^lv_{eE}) \frac{(\sigma^l)^2}{2} \\
& + ((k^h)^2v_{ee} + (K^h)^2v_{EE} + 2k^hK^hv_{eE}) \frac{(\sigma^h)^2}{2} \\
& \left. + [dc - (1 + \gamma)di] - \zeta v \right\} \\
\text{s.t. } & k_t^l + k_t^h \leq \Lambda e_t,
\end{aligned} \tag{12}$$

where we use sub-indices to denote partial derivatives and omit all function arguments for brevity. The left hand side of (12) reflects the fact that shareholders' required return equals ρ . The first line on the right hand side equals the change in shareholder value induced by a change in the bank's own capital, e , and the second line equals the change in value induced by a change in aggregate capital, E . The third and fourth lines capture the value impact of the variance of (and the covariance between) individual and aggregate bank capital. The first term in the fifth line reflects payments to the bank's shareholders net of capital injections and the second term reflects the bank's failure in case of a systemic crisis.

Individual banks' optimization problems can be greatly simplified by observing that the shareholder value function $v(e, E)$ is homogeneous of degree one in individual bank capital e . That is, when we multiply the initial condition e_0 by some factor $n > 0$, it is clearly optimal for banks to follow a strategy that consists of equivalently scaled controls ni, nc , and nk . Since both the feasible set of strategies and the objective function itself are thus homogeneous, the shareholder value in (10) satisfies

$$v(ne, E) = nv(e, E). \tag{13}$$

We define the scale-adjusted version of the bank's policies, for $n = 1/e$, which can be interpreted as the bank's seasoned offerings relative to outstanding equity di/e , its dividend-to-equity ratio dc/e , and its asset-to-equity ratios $a^l = k^l/e$ and $a^h = k^h/e$, the sum of which has to be lower than Λ by the capital requirement (8). Likewise, by using (13), define the bank's scaled shareholder value, or market-to-book ratio of equity, as

$$u(E) := v(1, E) = \frac{v(e, E)}{e}. \quad (14)$$

The market-to-book ratio of equity $u(E)$ is the same for all banks and a deterministic function of aggregate bank equity E . Hence, we may indeed focus on equilibria where bank loans only depend on aggregate equity, and not on its distribution among banks. We thus consider only Markov equilibria in which the loan rate is a deterministic function of aggregate bank capital (see Definition 1). This is equivalent to considering a “representative bank” that solves a scaled version of the original stochastic control problem with aggregate bank capital E as the single state variable.

Recapitalization and Dividend Policies. To characterize banks' equity issuance and payout policies, we take first order conditions in equation (12) and use the fact that by (14), the marginal value of equity is equal to the market-to-book ratio: $v_e(e, E) = u(E)$. It then follows immediately that it is optimal to make payments to shareholders ($dc > 0$) if and only if $u(E) \leq 1$. That is, as long as the value of maintaining book equity inside the bank is higher than shareholders' marginal value of receiving a payout, banks retain profits to build up equity buffers. Similarly, raising new capital ($di > 0$) is optimal if and only if $u(E) \geq 1 + \gamma$. That is, as long as the value of an additional unit of book equity is smaller than the marginal costs of raising new capital, only the internally generated equity is used to absorb losses on a bank's loans.

These considerations give rise to “barrier-type” payout and recapitalization strategies: Banks issue new equity if aggregate capital reaches a lower bound $\underline{E}$, which satisfies

$$u(\underline{E}) = 1 + \gamma. \quad (15)$$

Similarly, banks pay out earnings to shareholders if aggregate capital reaches an upper bound

$\bar{E}$, which is characterized by

$$u(\bar{E}) = 1. \quad (16)$$

Between the two boundaries, banks make no payments to shareholders and do not issue new capital.¹⁶ The payout boundary is pinned down by a standard no-arbitrage condition,

$$v(e - dc, \bar{E} - dC) + dc = v(e, \bar{E}). \quad (17)$$

That is, the ex-dividend equity value plus the dividend payment must equal the cum-dividend equity value. Applying a Taylor expansion to the left-hand side of (17), while using the homotheticity property $v(e, E) = eu(E)$, yields:

$$u(\bar{E})dc + eu'(\bar{E})dC = dc.$$

Since $u(\bar{E}) = 1$ by (16) and book equity e must be strictly positive by (11), it follows that

$$u'(\bar{E}) = 0. \quad (18)$$

A similar no-arbitrage condition has to hold at the issuance boundary:

$$v(e + di, \underline{E} + dI) - (1 + \gamma)di = v(e, \underline{E}),$$

which, after applying a Taylor expansion, yields

$$u(\underline{E})di + eu'(\underline{E})dI = (1 + \gamma)di. \quad (19)$$

Since $u(\underline{E}) = 1 + \gamma$ by the boundary condition (15) and the book equity e must remain strictly positive by (11), it follows that

$$u'(\underline{E}) = 0. \quad (20)$$

Since banks are homothetic, they all follow the same strategy, so that aggregate payments

¹⁶It is important to stress that, in contrast to the partial equilibrium models featuring similar barrier-type recapitalization and dividend policies (see e.g. Bolton et al., 2011; Hugonnier & Morellec, 2017), in our framework these boundaries are in terms of the aggregate, rather than the individual state.

($dC > 0$) cause aggregate capital to reflect at $\bar{E}$. Similarly, banks' joint issuance strategies prevent aggregate capital from falling below $\underline{E}$. In between the two boundaries, there are no capital injections or payouts, i.e.,

$$dC(E) = dc(E) = dI(E) = di(E) = 0 \text{ for } E \in (\underline{E}, \bar{E}) \quad (21)$$

Intuitively, when the banking system is well capitalized, the aggregate loan supply is high, making lending relatively unprofitable. Hence, banks pay out profits to shareholders at $E \geq \bar{E}$ (*payout region*). Banks issue new equity at $E \leq \underline{E}$ (*external financing region*), when the banking system is poorly capitalized and the aggregate loan supply is so low that lending is highly profitable. Between the boundaries, banks retain profits and convert them into book equity, which is used to absorb losses and finance loans to the real sector (*internal financing region*).

Equilibrium Loan Rate and Market-to-Book Value. Using the homotheticity property (14) and the fact that banks follow a barrier strategy, we rewrite equation (12) for the *internal financing region* $(\underline{E}, \bar{E})$ as:

$$\begin{aligned} \zeta u(E) = & \left[E\rho + (R^l(E) - \rho)K^l(E) + (R^h(E) - \rho)K^h(E) \right] u'(E) \\ & + \left[(\sigma^l K^l(E))^2 + (\sigma^h K^h(E))^2 \right] \frac{u''(E)}{2} \\ & + \max_{a^l} a^l \left[\underbrace{(R^l(E) - \rho)u(E) + (\sigma^l)^2 K^l(E)u'(E)}_{=: \eta^l(E)} \right] \\ & + \max_{a^h} a^h \left[\underbrace{(R^h(E) - \rho)u(E) + (\sigma^h)^2 K^h(E)u'(E)}_{=: \eta^h(E)} \right] \\ \text{s.t. } & a^l + a^h \leq \Lambda. \end{aligned} \quad (22)$$

In equilibrium, bank equity must earn an expected excess return equal to the crisis intensity ζ : the term $\zeta u(E)$ on the left-hand side of (22) is the compensation shareholders require for the risk that a systemic crisis wipes out their claim. Since (22) is a second-order ODE, it requires two boundary conditions, (15) and (16), to find a solution. The two free boundaries, $\underline{E}$ and $\bar{E}$, are determined by the no-arbitrage conditions (18) and (20).

Now consider the equilibrium asset-to-equity ratios a^i (or equivalently, bank lending $k^i = a^i \cdot e$) for the two types of corporate borrowers $i \in \{l, h\}$. Note that the HJB (22) is linear in a^i . Hence, the minimum capital requirement is binding if at least one of the marginal returns on lending η^i is strictly positive. The bank leverage is then fully determined by the binding condition (8), i.e., $a^l + a^h = \Lambda$. If the constraint binds for an individual bank, it binds also on the aggregate level due to homotheticity. Using the market clearing condition $K^i = L^i(R^i)$, the equilibrium loan rates thus have to satisfy

$$L^l(R^l(E)) + L^h(R^h(E)) = \Lambda E. \quad (23)$$

The total amount of lending is thus given by the binding minimum capital requirement, and lending in both markets increases with aggregate capital. Furthermore, the equilibrium loan rates R^l and R^h have to satisfy the individual rationality constraint of the banks. That is, for banks' individual lending policies to be compatible with the aggregate equilibrium levels, they may not benefit from reallocating funds from one loan market to the other, as $\eta^h = \eta^l > 0$, or

$$\frac{R^h(E) - R^l(E)}{L^h(R^h(E))(\sigma^h)^2 - L^l(R^l(E))(\sigma^l)^2} = -\frac{u'(E)}{u(E)} > 0, \quad (24)$$

for $E \in [0, E^*]$.

The terms η^l and η^h in (22) can be interpreted as the “shadow costs” associated with the capital requirement. In particular, a binding capital requirement is associated with a *higher* market-to-book ratio, that is, $\eta^i(E) > 0$. This seemingly counterintuitive result reflects the fact that a binding capital requirement restricts aggregate loan supply, so that the equilibrium loan rates increase and lending becomes more profitable.

For the total loan volume to be interior in equilibrium (i.e. the capital requirement to be slack), both shadow costs attributed to the capital requirement η^l and η^h must be zero. This ensures that banks do not have an incentive to increase lending in either market above the equilibrium level. Thus, setting $\eta^l = \eta^h = 0$ yields the following results.

$$\frac{R^l(E) - \rho}{L^l(R^l(E))(\sigma^l)^2} = \frac{R^h(E) - \rho}{L^h(R^h(E))(\sigma^h)^2} = -\frac{u'(E)}{u(E)}, \quad (25)$$

for the region where the capital requirement is slack.¹⁷ Since the market-to-book ratio of bank capital is decreasing in aggregate bank capital (as we show in the Proof of Proposition 1), the condition (25) implies that banks require a positive spread over the common discount rate ρ . This reflects the following mechanism: When an individual bank makes profits, all other banks also make profits. As the banking sector thus becomes better capitalized, the aggregate loan supply increases, which lowers the equilibrium loan rate. With lower lending profits, the market-to-book ratio $u(E)$ decreases. Similarly, when a bank makes losses, other banks also make losses. As the capitalization of the banking system deteriorates, this reduces the aggregate loan supply, and the market-to-book ratio of equity increases. This effect is similar to the decreasing marginal utility of consumption for a risk-averse individual.

Proposition 1. *A competitive Markov equilibrium is characterized by the following properties.¹⁸*

1. *The loan rates $R^l(E)$ and $R^h(E)$ are given by (23) and (24) in the region $E \in [\underline{E}, E^*)$, and by (25) in the region $E \in [E^*, \overline{E}]$.*
2. *The critical capitalization E^* , at which capital requirement (8) becomes slack, is determined by continuity of $R^l(E)$ and $R^h(E)$. The requirement can bind only for $E \leq (L^l(\rho) + L^h(\rho))/\Lambda$.*
3. *The equilibrium market-to-book ratio $u(E)$ satisfies the Hamilton-Jacobi-Bellman equation (22) subject to the boundary conditions (15), (16), (18), and (20).*

¹⁷Note that the equilibrium spreads, $R^i(E) - \rho$, in (25) can also be interpreted as “hedging premia,” rather than standard risk-premia: They contain the *cross derivative* of the bank’s shareholder value with respect to individual and aggregate equity, i.e., $v_{eE}(e, E)/v_e(e, E) = u'(E)/u(E)$, and not its second derivative, as a measure of risk aversion. A similar mechanism drives the intertemporal hedging demand in the dynamic model of liquidity provision by Kondor and Vayanos (2019).

¹⁸The problem of finding $u(E)$ together with the free boundaries $\underline{E}, \overline{E}$ is a singular-control problem of the barrier type studied in the corporate-liquidity and bank-capital literature on which our model builds, where existence and uniqueness of the value function under proportional issuance costs and an endogenous payout boundary are established; see Décamps et al. (2011) and Hugonnier and Morellec (2017). The specificity of our setting is that the loan rates $R^i(E)$ are determined in equilibrium, which couples this free-boundary problem to the market-clearing conditions (23) and (25). We resolve the coupled system numerically; the proof reduces it to a one-dimensional shooting problem in the issuance boundary $\underline{E}$.

4. Aggregate bank capital evolves according to

$$\begin{aligned} dE_t = & E_t \rho dt + L^l(R^l(E_t)) [(R^l(E_t) - \rho)dt - \sigma^l dZ_t^l - \phi dN_t] \\ & + L^h(R^h(E_t)) [(R^h(E_t) - \rho)dt - \sigma^h dZ_t^h - \phi dN_t], \end{aligned} \quad (26)$$

for $E_t \in (\underline{E}, \bar{E})$. It is reflected at the boundaries $\underline{E}$ and $\bar{E}$, and restarts from $\underline{E}$ after each crisis.

Note that all equilibrium objects are indeed deterministic functions of aggregate bank capital E , the single state variable. Similarly, the market value of a unit of individual book equity (the market-to-book ratio) $u(E)$, is also a deterministic function of aggregate capital. The equilibrium evolution of aggregate bank capital follows from the individual issuance and payout policies of the banks in (21), which in turn are individually optimal under the boundary conditions (15), (16), (18) and (20). That is, the absence of arbitrage on the equity market implies that individual banks cannot increase their shareholder value by deviating from the barrier strategy to raise new equity ($di > 0$) in $\underline{E}$ and to distribute dividends ($dc > 0$) at $\bar{E}$.

4 Equilibrium Analysis

Market Outcome for a Given Regulatory Regime. Consider first the equilibrium loan rates $R^i(E)$ and the aggregate lending $L^i(R^i)$. When aggregate capital is low (i.e., for $E \leq E^*$), leverage is fully determined by the binding capital requirement, and hence lending is strictly increasing in aggregate equity. Similarly, aggregate lending in both markets increases in aggregate equity also in the region where the constraint is slack. We derive the following auxiliary result showing that the equilibrium loan rates are strictly decreasing in aggregate equity when the constraint is slack.

Corollary 1. *In the equilibrium characterized by Proposition 1, the loan rates satisfy first-order differential equations:*

$$\frac{dR^i}{dE} = -\psi(E) \frac{L^i(R^i(E))(R^i(E) - \rho)}{L^i(R^i(E)) - \frac{dL^i}{dR^i}(R^i(E) - \rho)} < 0 \quad (27)$$

for $E \in [E^*, \bar{E})$ and $i \in \{l, h\}$, subject to the boundary condition:

$$R^i(\bar{E}) = \rho. \quad (28)$$

The expression $\psi(E) > 0$ is defined in the proof. At $\bar{E}$, equation (27) extends in the limiting sense, with a finite, strictly negative one-sided slope.

Calibration. All parameters are anchored in the empirical credit-risk and corporate-finance literature; Table 1 lists the values and sources. The crisis intensity $\zeta = 0.04$ corresponds to the two systemic banking crises in the United States (1988, a borderline case, and 2007) over 1970–2017 in the database of Laeven and Valencia (2020). The crisis loss $\phi = 0.18$ is set somewhat below the resolution losses on failed US banks over 2008–2013, a median of 23.6% of assets (Davison, 2020), because failed banks are the worst tail of the loss distribution, whereas ϕ applies to the assets of the entire banking sector.¹⁹ The loan-portfolio volatilities $(\sigma^l, \sigma^h) = (0.03, 0.10)$ follow the annualized volatility of credit excess returns on U.S. investment-grade and high-yield corporate bonds, 3.9% and 9.6% over 1988–2014 (Asvanunt & Richardson, 2017), with σ^l slightly below the investment-grade figure because that index includes BBB-rated issuers. The loan-demand elasticities $\beta^l = 5$ and $\beta^h = 4$ imply that a 100-basis-point increase in the loan rate lowers demand by 5% and 4% at the normalization $L^i(\rho) = 1$, between the semi-elasticities estimated on bank-level and credit-register data, about 1.4% (Wang et al., 2022) and 7% (at their mean loan rate; Crawford et al., 2018), and above the household-mortgage benchmark of 2–3% (DeFusco & Paciorek, 2017). The social-cost weight $\theta = 0.50$ in (7) combines the marginal cost of public funds, about 0.3 per unit of revenue in the United States (Ballard et al., 1985), with a loading for the large and persistent losses that crises impose on third parties (Cerra & Saxena, 2008; Laeven & Valencia, 2020).

The remaining parameters take standard values. The risk-free rate $\rho = 0.05$ lies somewhat above the average one-month Treasury-bill rate of 4.1% over 1954–2024.²⁰ The issuance cost $\gamma = 0.2$ lies between the direct flotation costs of seasoned equity offerings, about 5% of the

¹⁹The calibrated value also satisfies the model’s requirement that a crisis wipes out bank equity at every reachable capitalization, $\phi \geq \max_t \bar{E}_t/2 = 0.164$, since aggregate lending equals two at the payout boundary under the normalization $L^i(\rho) = 1$.

²⁰Computed from the risk-free rate series in the Kenneth R. French Data Library.

proceeds (see the estimates collected in Décamps et al., 2011, p. 1508; for banks, Boyson et al., 2016, Table 1, Panel C), and their sum with the announcement-day loss in firm value of 31% of the proceeds for primary offerings by industrial firms (Asquith & Mullins, 1986, Table 3). Both components are transfers from shareholders, to underwriters and to new investors, rather than resource costs, so that γ shapes banks' policies but does not enter the welfare criterion of Section 6. The loan-demand intercepts $\alpha^l = 1.25$ and $\alpha^h = 1.20$ pin down the upper bound of the firm-productivity distribution at $\alpha^i/\beta^i = 25\%$ and 30% ; see Appendix B for the microfoundation.

The model is a stylized illustration of the anticipation channel rather than a moment-matched calibration, but the implied leverage is in a plausible range. In the old regime, the lending-to-equity ratio equals 7.5 at the payout boundary $\bar{E}_{old}$, the deterministic steady state, and about 13 on average under the stationary distribution (median 11), which accounts for the losses that push capital back towards the binding region; on this distinction, see Klimenko et al. (2017). The two values bracket the assets-to-equity ratio of about 9 for US commercial banks over 2012–2019 (FDIC Historical Bank Data); in the binding region the ratio equals $\Lambda_{old} = 25$ by construction.

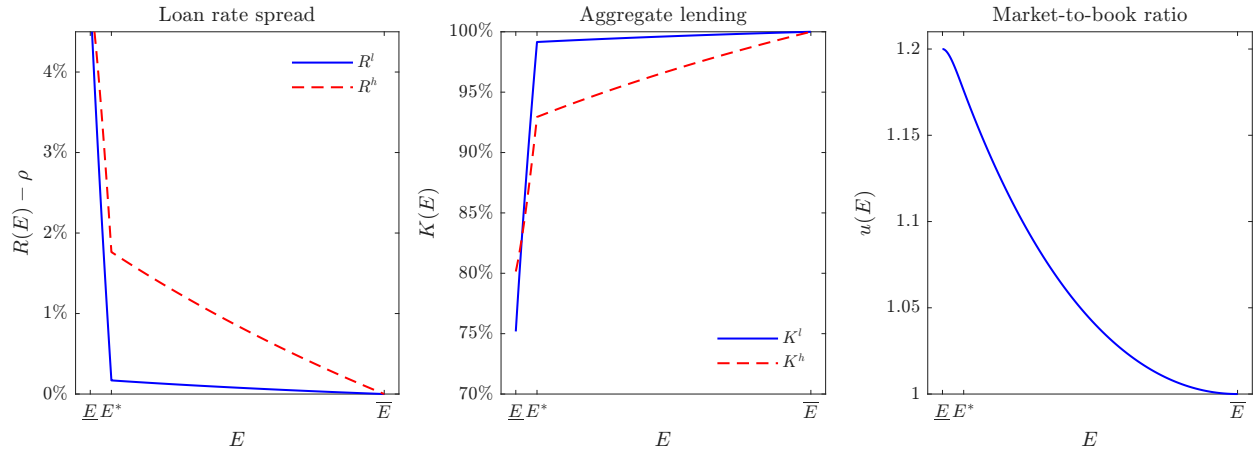


Figure 1: Illustrates the equilibrium loan rate, aggregate lending, and the market-to-book ratio of equity as a function of aggregate capital E , for a maximum leverage of $\Lambda_{old} = 25$ (which corresponds to a minimum regulatory capital ratio of 4%). Aggregate lending is expressed relative to the level of aggregate lending in the friction-less benchmark. Parameter values per Table 1.

Figure 1 illustrates the equilibrium characterized in Corollary 1 for these parameter values. Boundary condition (28) has an intuitive interpretation: At the payout boundary

Table 1: Calibration of the numerical illustrations.

Parameter	Symbol	Value	Anchor
Empirical credit-risk anchors			
Crisis intensity	ζ	0.04	2 US crises, 1970–2017 (Laeven & Valencia, 2020)
Loss given default	ϕ	0.18	Below crisis failed-bank loss rates (Davison, 2020)
Low-risk loss volatility	σ^l	0.03	IG credit excess returns (Asvanunt & Richardson, 2017)
High-risk loss volatility	σ^h	0.10	HY credit excess returns (Asvanunt & Richardson, 2017)
Loan-demand parameters (microfoundation: Appendix B)			
Low-risk demand elasticity	β^l	5	Loan-demand rate sensitivity (Crawford et al., 2018; DeFusco & Paciorek, 2017; Wang et al., 2022)
High-risk demand elasticity	β^h	4	Set below β^l
Low-risk demand intercept	α^l	1.25	Productivity ceiling $\alpha^l/\beta^l = 25\%$
High-risk demand intercept	α^h	1.20	Productivity ceiling $\alpha^h/\beta^h = 30\%$
Structural parameters			
Risk-free rate	ρ	0.05	Above 1954–2024 T-bill average (4.1%)
Issuance cost	γ	0.20	SEO cost: direct $\approx 5\%$, announcement loss 31% (Asquith & Mullins, 1986)
Regulation			
Old leverage limit	Λ_{old}	25	Tier 1 minimum under Basel II (4% of risk-weighted assets) (Basel Committee on Banking Supervision, 2006)
New leverage limit	Λ_{new}	12.5	Close to the Basel III Tier 1 minimum plus conservation buffer (6% + 2.5%) (Basel Committee on Banking Supervision, 2011)
Welfare			
Social-cost weight (per unit of bailout shortfall)	θ	0.50	MCPF ≈ 0.3 (Ballard et al., 1985) + third-party loading

$\bar{E}$, aggregate loan supply is so high that loan making becomes unprofitable at the margin. As illustrated in the left panel of Figure 1, the loan rate in each market is in fact highest in the external financing region $E \leq \underline{E}$ and achieves its minimum in the payout region $E \geq \bar{E}$. Aggregate lending in both loan markets is thus strictly increasing in E (see middle panel of Figure 1). Furthermore, as illustrated in the right panel of Figure 1, the market-to-book ratio of bank equity obtains its maximum of $1 + \gamma$ in the external financing region, $E \leq \underline{E}$, where lending is so profitable that banks raise new capital to avoid further unlevering in the face of a binding capital requirement. The market-to-book ratio is strictly decreasing in E and obtains its minimum in the payout region $E \geq \bar{E}$, where loan making becomes unprofitable at the margin, as $R^i(\bar{E}) = \rho$. These testable implications are summarized in the following Corollary:

Corollary 2. *In the competitive Markov equilibrium characterized in Proposition 1, it holds that*

- (i) *the loan rates, $R^l(E)$ and $R^h(E)$, are both strictly decreasing in aggregate bank capital;*
- (ii) *aggregate loan volumes, $K^l(E)$ and $K^h(E)$, are strictly increasing in aggregate bank capital;*
- (iii) *the market-to-book ratio of equity, $u(E)$, is strictly decreasing in aggregate bank capital.*

Outcome under Different Regulatory Regimes. Before analyzing the implementation strategy for regulatory tightening, we briefly compare the respective market outcomes under two different regulatory regimes. We refer to the milder capital requirements as the “old regulatory regime” (Λ_{old}) and to the stricter requirements as the “new regulatory regime” (Λ_{new}). The analysis in this section can thus be interpreted as an unanticipated regulatory tightening without any transition phase.

For the issuance boundary $\underline{E}$, the effect is straightforward: Since the capital requirement binds at $E = \underline{E}_{old}$, a reduction of the maximum leverage reduces the total supply of loans and, thus, increases the equilibrium loan rates. This makes lending more profitable, which would, all else equal, increase the market-to-book ratio to a value $u_{new}(\underline{E}_{old}) > 1 + \gamma$. By continuity, the new issuance boundary $\underline{E}_{new}$ thus has to be higher than $\underline{E}_{old}$. The same reasoning applies to the critical level of aggregate capital where the capital requirement

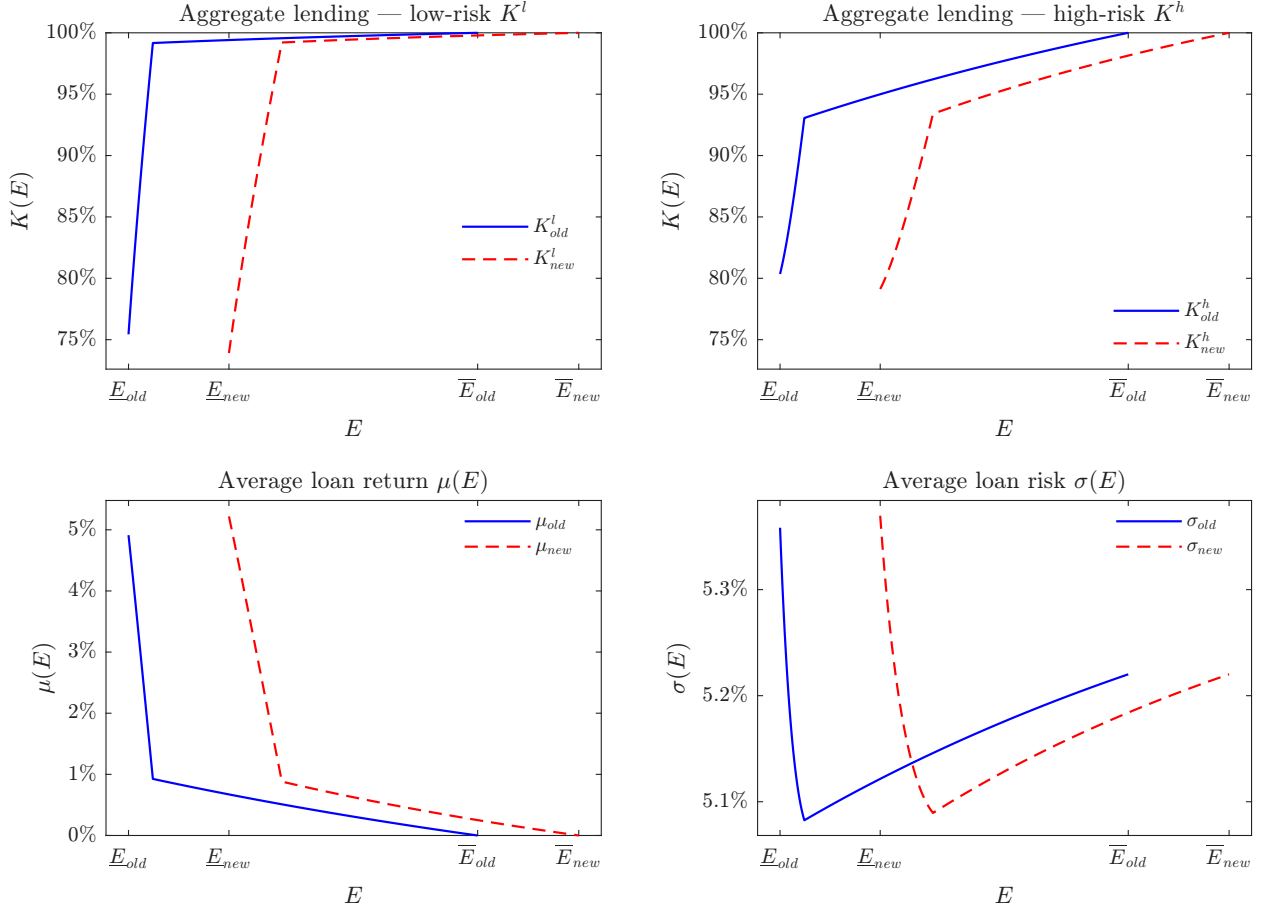


Figure 2: Stationary-equilibrium comparison of the two regulatory regimes $\Lambda_{old} = 25$ and $\Lambda_{new} = 12.5$. The top row shows aggregate lending in the low-risk segment $K^l(E)$ and the high-risk segment $K^h(E)$; the bottom row shows the average loan return $\mu(E)$ and average loan risk $\sigma(E)$. Solid blue lines: old regime; red dashed lines: new regime. Parameter values per Table 1.

becomes slack, i.e., $E_{new}^* > E_{old}^*$. This implies that at any given level of capital, the tighter regulation Λ_{new} is more likely to *become* binding than the milder regulation Λ_{old} . This increases the market-to-book ratio of equity also for levels of aggregate bank capital where the constraint is slack. Hence, we would have $u_{new}(\bar{E}_{old}) > 1$, which implies that the payout boundary has to increase as well: $\bar{E}_{new} > \bar{E}_{old}$.²¹

²¹Note that ceteris paribus, a tighter regulation would lead to more frequent recapitalization, which tends to decrease the market-to-book ratio at a given level of aggregate capital and, thus, decrease the payout and issuance boundaries. However, in our numerical analysis this countervailing effect dominates only for excessively high levels of regulation under which the constraint is binding for all $E \in [\underline{E}, \bar{E}]$. In the

Furthermore, unanticipated regulatory tightening without a transition phase can lead to severe losses in aggregate lending. Figure 2 illustrates that, in the region where the old constraint is slack but the new constraint just begins to bind, aggregate lending to low-risk borrowers can drop by about 25% (top-left panel) and lending to high-risk borrowers by about 17% (top-right panel) in the numerical example considered. Note that the most severe losses in lending occur at levels of capital where the mild regulation Λ_{old} is slack, but the tighter regulation Λ_{new} becomes binding if it is introduced (i.e., for $E \in [E_{old}^*, \underline{E}_{new}]$). At the same time, an unexpected regulatory tightening triggers an immediate capital injection of $(\underline{E}_{new} - E)$ if $E < \underline{E}_{new}$, which reduces the social costs in case of a banking crisis.

This discussion suggests that a transition phase might indeed improve the trade-off between financial stability and losses in lending. Therefore, we next analyze how a (longer) transition phase affects the dynamics of capital accumulation and aggregate lending.

5 Regulatory Tightening with a Transition Phase

We have seen in the last section that a sudden regulatory tightening may trigger immediate capital injections but has the potential to cause a severe credit crunch. In line with common regulatory practice, we, therefore, now consider the introduction of the regulatory tightening over a transition phase.

5.1 Equilibrium in the Transition Phase

In $t = 0$, the regulator announces a new capital requirement Λ_{new} , to which all banks must adhere after a transition phase of T years. Throughout the transition phase the requirement in force remains Λ_{old} , so that T measures the delay between the announcement of the reform and its implementation: what is gradual is banks' adjustment, not the constraint itself. We nonetheless refer to T as the length of the transition phase.²² The market-to-book ratio of

region where the constraint is slack, however, this effect is offset by the fact that a higher probability of recapitalization increases banks' implied risk-aversion and, hence, the required lending premium. Higher equilibrium loan rates, however, reduce banks' exposure and thus the frequency of recapitalization.

²²In earlier versions of the model we also solved a continuous tightening of the requirement over the transition phase, in which the anticipation mechanism operates in the same way, and a transition of uncertain length that ends at a constant Poisson rate.

banks $u(E, t)$ during the transition phase satisfies the following Hamilton-Jacobi-Bellman equation:

$$\begin{aligned}
\zeta u(E, t) = & \left[E\rho + (R^l(E, t) - \rho)K^l(E, t) + (R^h(E, t) - \rho)K^h(E, t) \right] u_E(E, t) \\
& + \left[(\sigma^l K^l(E, t))^2 + (\sigma^h K^h(E, t))^2 \right] \frac{u_{EE}(E, t)}{2} + u_t(E, t) \\
& + \max_{a_t^l} a_t^l \left[(R^l(E, t) - \rho)u(E, t) + (\sigma^l)^2 K^l(E, t)u_E(E, t) \right] \\
& + \max_{a_t^h} a_t^h \left[(R^h(E, t) - \rho)u(E, t) + (\sigma^h)^2 K^h(E, t)u_E(E, t) \right] \\
\text{s.t. } & a_t^l + a_t^h \leq \Lambda_{old}.
\end{aligned} \tag{29}$$

subject to the usual boundary conditions

$$u(\underline{E}_t, t) - (1 + \gamma) = u(\overline{E}_t, t) - 1 = u_E(\underline{E}_t, t) = u_E(\overline{E}_t, t) = 0.$$

Compared to (22), HJB (29) is now a partial differential equation with calendar time as a second state variable. Hence, also the free boundaries are now functions of time. In the time dimension, the boundary condition reflects the fact that at the end of the transition phase, where no further regulatory changes are expected, the solution to (29) is equal to the stationary solution to the problem considered in Section 4 under tighter capital requirements Λ_{new} :

$$u(E, T) = u_{new}(E).$$

The left panel of Figure 3 illustrates the market-to-book ratio for the mild capital requirement ($u_{old}(E)$, the blue solid line), for the strict capital requirement ($u_{new}(E)$, the red dashed line) and for a point in time halfway in the transition phase ($u_{tr}(E) := u(E, t)$ with $t = T/2$, the black dash-dotted line; below, the subscript tr refers to this point in time for loan rates, lending, and the distribution of social costs as well). It shows that for high levels of aggregate capital, the market-to-book ratio in the transition phase u_{tr} approaches u_{new} . Hence, the anticipated increase in the profitability of lending is fully “priced-in” in the region where the capital requirement is far from binding. For low levels of aggregate capital, on the contrary, the market-to-book ratio is almost unaffected by the regulatory announcement, as

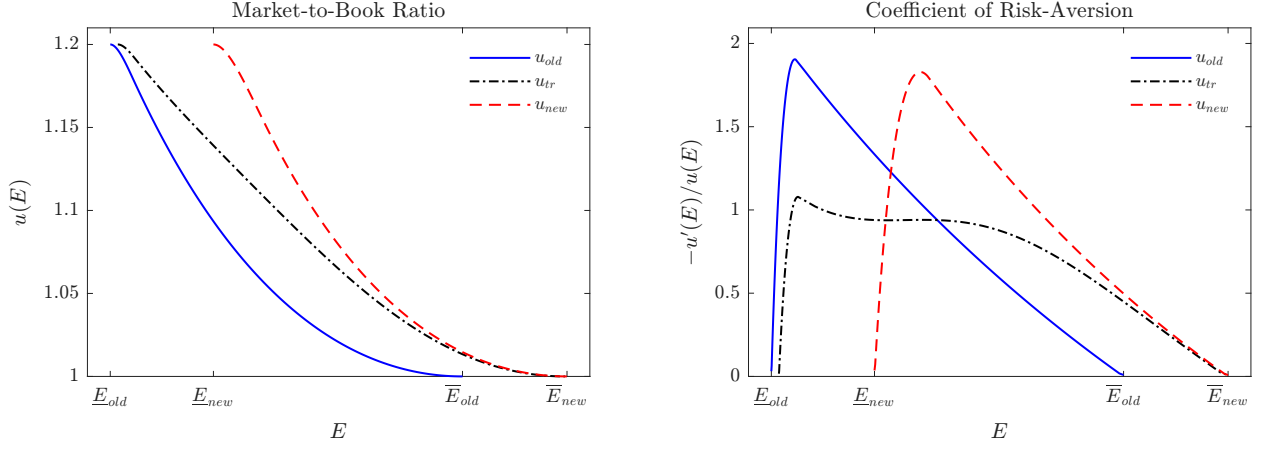


Figure 3: Illustrates the market-to-book ratio function and banks' implied coefficient of risk-aversion before, during and after the transition phase for a regulatory change from $\Lambda_{old} = 25$ to $\Lambda_{new} = 12.5$. The length of the transition phase equals one year. Parameter values per Table 1.

in this region, the total amount of lending is determined by the binding capital requirement. Hence, since the market-to-book ratio $u_{tr}(E)$ is biased downward in this region, it falls more slowly in aggregate capital, in the region where the constraint becomes slack. Intuitively, as aggregate capital increases, the system moves closer to the states of the world, in which capital is already more valuable due to the anticipated regulatory tightening. The right panel of Figure 3 shows that this decrease in $|u'_{tr}(E)|$ implies that banks become less “risk-averse” in response to the regulatory announcement. That is, their implied coefficient of risk aversion $-u'_{tr}(E)/u_{tr}(E)$ (the black dashed-dotted line) decreases, which in turn leads to lower equilibrium loan rates $R_{tr}^i(E)$, as specified in (25). Quite surprisingly, the anticipation of a future *scarcity* of loans therefore *reduces* the current loan rates.

The effect of a regulatory announcement on aggregate lending functions is illustrated in the top row of Figure 4, in the left panel for low-risk loans $K^l(E)$ and in the middle panel for high-risk loans $K^h(E)$. Recall that during the transition phase, the old capital requirement applies. Hence, when this requirement is binding, the announcement of a regulatory tightening in T has no effect on aggregate lending. Recall from Figure 3 that the anticipated regulatory change is, by contrast, fully priced in for high levels of capital. Hence, K_{tr}^i approaches K_{new}^i in this region, such that the announcement of tighter regulation leads to a reduction in lending in this region. However, for intermediate levels of aggregate capital, the anticipation of regulatory tightening has the opposite effect. Recall from Figure 3 that the

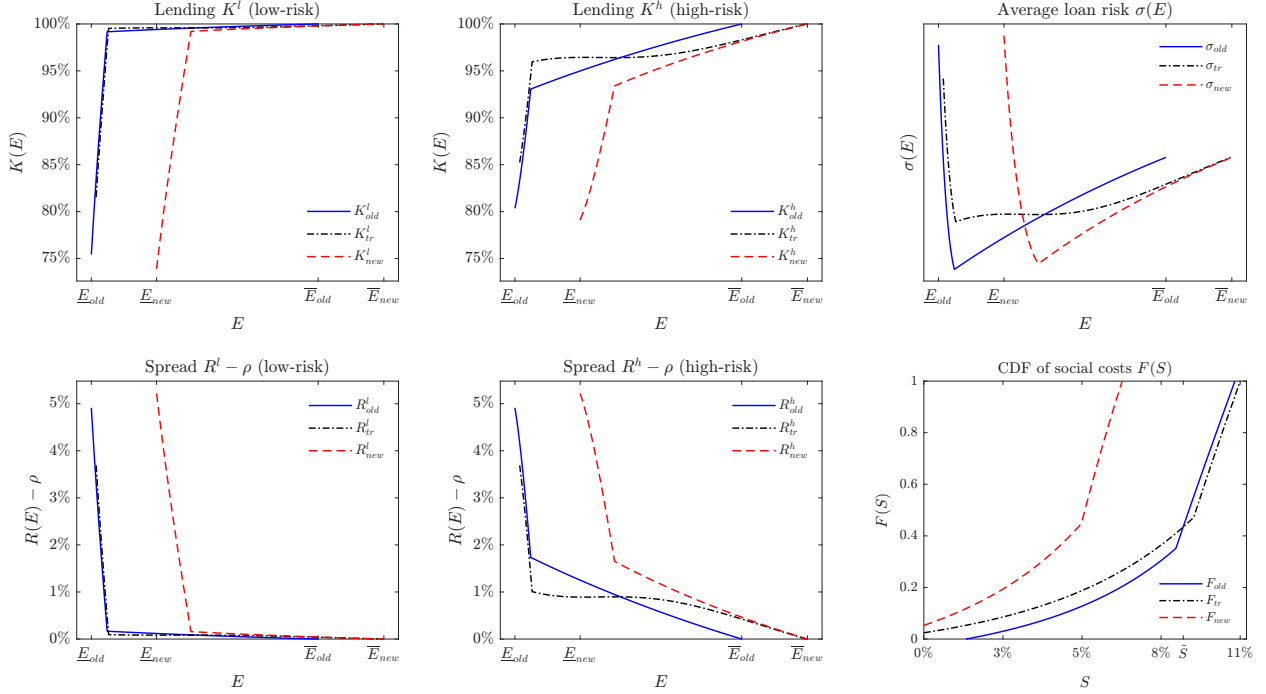


Figure 4: Illustrates the aggregate lending functions (top row, left/middle: low-risk K^l and high-risk K^h), loan rate spreads (bottom row, left/middle: $R^l - \rho$ and $R^h - \rho$), average loan risk $\sigma(E)$ (top right), and the CDF of social costs of crises (bottom right). The *blue solid* lines refer to a point in time before the announcement of the regulatory change from $\Lambda_{old} = 25$ to $\Lambda_{new} = 12.5$; the *red dashed* lines to a point in time after its implementation; and the *black dash-dotted* lines to the point in time halfway through the transition phase. The length of the transition phase equals 1 year. Values are expressed relative to aggregate lending in first-best. Parameter values per Table 1.

announcement of a regulatory tightening decreases the implied coefficient of risk aversion of banks in this region, such that the equilibrium loan rates decrease below R_{old}^i (see the bottom row of Figure 4). Hence, aggregate lending increases above K_{old}^i (see the top row of Figure 4). This is in line with recent empirical evidence that aggregate bank lending increased after the announcement of Basel III (see Basel Committee on Banking Supervision, 2022, Section 8.1.3 and Annex A.18) and our own empirical analysis in Section 5.2.

The upper right panel of Figure 4 shows the average riskiness of the loan portfolio, measured as the diffusion coefficient of aggregate bank capital per unit of lending,

$$\sigma(E) := \frac{[(\sigma^l K^l(E))^2 + (\sigma^h K^h(E))^2]^{1/2}}{K^l(E) + K^h(E)},$$

where the sum of squares reflects the independence of the Brownian shocks across the two loan segments. Note that for intermediate levels of aggregate capital, the average loan becomes more risky in response to the regulatory announcement. That is, while lending in both markets increases in response to the announcement of a regulatory tightening, the increase in lending is more pronounced in the high-risk loan market, as banks rebalance their portfolio towards more risky loans.²³ This is intuitive, as banks become effectively less risk-averse in this region (see Figure 3).

Finally, the lower right panel illustrates the cumulative distribution function of the social costs S of a crisis, as defined in (7), computed from the stationary distribution of aggregate capital that the respective policy functions imply. The social costs in the case of a systemic crisis under the old regulatory regime first-order stochastically dominate those under the new regulatory regime. Hence, as stricter capital requirements lead to higher levels of capital and lower levels of lending, they tend to reduce the shortfall that the regulator has to cover in the case of a systemic banking crisis.

During the transition phase, this ranking breaks down. For high levels of capital, the announcement raises the payout boundary and lowers lending, so that banks' exposure falls and probability mass shifts to the lower tail of the distribution of social costs. For low levels of capital, the binding requirement pins down lending and the issuance boundary, so that the announcement changes nothing. For intermediate levels of capital, however, lending rises in response to the announcement, and so does the exposure $\phi(K^l + K^h) - E$, which shifts probability mass to the upper tail. The announcement thus rotates the CDF of social costs clockwise around a critical level $\tilde{S}$, such that $F_{tr}(S) > F_{old}(S)$ for $S < \tilde{S}$ and $F_{tr}(S) < F_{old}(S)$ for $S > \tilde{S}$. Intuitively, the announcement makes the banking sector safer where it is already well capitalized, because banks build buffers ahead of the tightening, and riskier where it is only moderately capitalized, because the same anticipation lowers their effective risk aversion. Rather than lowering the social cost of a crisis across the board, the announcement spreads it out; whether it lowers the expected cost of crises during the transition depends on the initial capitalization and on the length of the transition, which the welfare analysis in Section 6 takes into account.

²³The shift arises where the requirement in force is slack, so it is not driven by the two loan types carrying the same risk weight in (8): the pricing condition (25) that governs the composition of lending does not involve the requirement.

5.2 Evidence

This section evaluates the empirical implications of our model using data on bank lending, loan rates, and regulatory capital for a large unbalanced panel of U.S. Bank Holding Companies (BHCs) over the period 2005–2017, obtained from the Worldscape database via Datastream²⁴.

The global announcement of the Basel III reform occurred on December 16, 2010, while domestic implementation rules were announced later in most jurisdictions. In the United States, the jurisdictional announcement date was August 30, 2012. According to Basel Committee on Banking Supervision (2022), full implementation following the jurisdictional announcement took, on average, five years across jurisdictions. We therefore define the transition period as 2012–2017.

Figure 5 displays the evolution of aggregate lending and (weighted) averages of loan rates over the sample period. The vertical lines indicate the 2010 global announcement and the 2012 U.S. jurisdictional announcement of Basel III. During the transition period, total loans increased while lending rates declined. This pattern suggests that the phased introduction of tighter capital regulation may have supported credit supply and reduced loan rates, consistent with the international evidence reported in Basel Committee on Banking Supervision (2022) and with the implication of our model.

The comparisons that follow are simple difference-in-difference style statistics that contrast average lending and loan rates before and after the Basel III jurisdictional announcement across bank groups. These comparisons implicitly assume that macroeconomic developments over 2012–2017, such as the post-crisis recovery and the decline in interest rates associated with accommodative monetary policy, shifted outcomes in parallel across groups. Under this assumption, the differences in difference of averages across groups provide a reduced-form measure of the heterogeneous response to the regulatory phasing-in. While these statistics do not constitute a fully identified causal design, they offer a simple and informative way to relate the model’s qualitative implications to observed cross-sectional patterns in lending and loan pricing during the transition period.

²⁴The bank variables correspond to the following Worldscape data points: (log) total loans (WC02271); the lending rate, measured by the ratio of interest and fees on loans (WC01007) to total loans; the regulatory capital ratio is item WC18158.

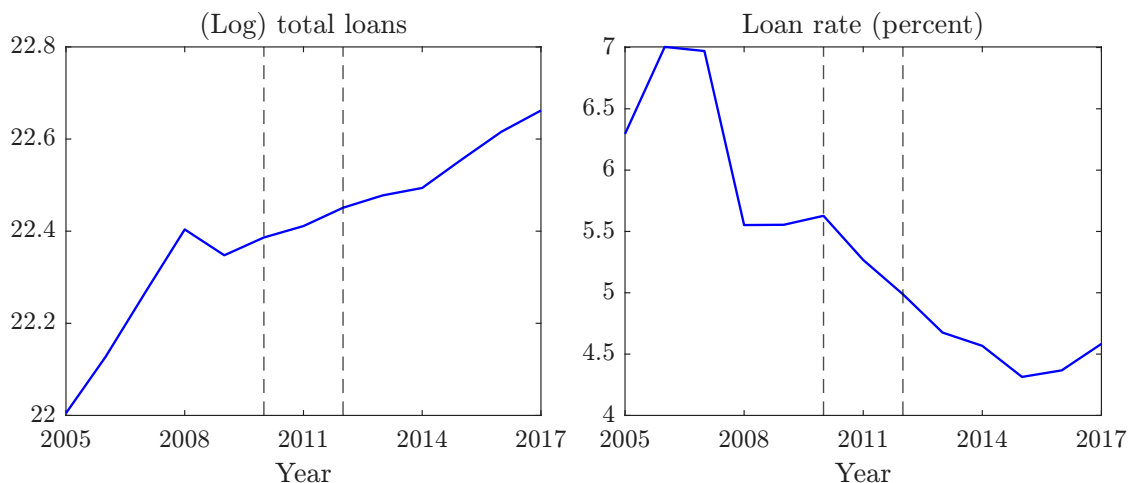


Figure 5: U.S. Bank Holding Companies: Aggregate Lending and Loan Rates. Log of total loans summed over the banks in the panel (left) and the loan-weighted average loan rate, interest and fees on loans over total loans, in percent (right), 2005–2017. The vertical lines mark the global announcement of Basel III in December 2010 and the U.S. jurisdictional announcement in August 2012.

Capital Heterogeneity. As shown in Section 5.1, the model predicts that during the phasing-in of higher regulatory capital requirements, banks with *intermediate* capitalization, defined as capital levels above the binding requirement but below levels substantially exceeding it, should increase lending and reduce loan rates. Although the model is a representative-bank general equilibrium with a single aggregate state, the cross-sectional dispersion of bank capitalization in the data offers a natural setting in which this mechanism should be visible: banks in the intermediate-capital region should display higher changes in lending and stronger declines in loan rates than those of banks far above or below that region.

Evidence consistent with this qualitative implication appears in Table 2, which reports average lending and loan rates for a fixed sample of 490 banks, classified by their 2011 regulatory capital ratio and followed over 2005–2017 in every year in which regulatory capital is reported: “low” (below 11%, about 4% of the sample), “intermediate” (between 11% and 24%, about 88% of the sample), and “high” (above 24%, about 8% of the sample). The 11% lower cutoff corresponds to capital levels around the combined Basel III requirements (CET1, Tier 2 minimum, plus the conservation and counter-cyclical buffers, and systemically important bank requirements) and typical supervisory expectations (roughly 10–11% for U.S. BHCs over the sample period), while the 24% upper cutoff identifies banks with a large

buffer above the combined Basel III requirements, corresponding to the 95th percentile of the distribution of regulatory capital over 2005–2017.

Table 2: Average Lending and Loan Rates by Capitalization Group

(1)		(2)	(3)	(4)	(5)	(6)	(7)
Regulatory capital groups	Banks	2005–2011		2012–2017		Δ	$\Delta\%$
		(log) loans	loan rate	(log) loans	loan rate	(log) loans	loan rate
Low (< 11%)	22	12.91 (0.21)	6.43 (0.10)	13.36 (0.33)	5.07 (0.15)	0.45 (0.25)	–21.15
Intermediate (11–24%)	431	13.74 (0.08)	6.25 (0.04)	14.44 (0.08)	4.67 (0.04)	0.70 (0.04)	–25.23
High (> 24%)	37	13.29 (0.22)	6.13 (0.14)	13.95 (0.18)	4.73 (0.21)	0.66 (0.11)	–22.76

Note. Banks are assigned to a capitalization group on the basis of their 2011 regulatory capital ratio (Worldscope WC18158) and retain that assignment in every year in which regulatory capital is reported. Columns (2)–(5) are averages of (log) total loans and of the loan rate, in percent, over the pre- and post-announcement periods. Column (6) is the difference in average (log) loans; column (7) is the percentage change in the average loan rate. Standard errors in parentheses are clustered by bank. For the Low, Intermediate, and High groups, the standard errors for level changes in the loan rate are 0.160, 0.030, and 0.174, respectively; none of the differences in changes across groups is significant at conventional levels. Sample construction: loan rates above 28 percent (15 bank-years) and regulatory capital ratios that are non-positive, at or above 100 percent, or in the top percentile of the pooled 2005–2017 distribution (63 bank-years in total) are treated as coding errors and set to missing before classification; assignment to a group in 2011 requires non-missing regulatory capital, loans, and loan rate in that year. The loan columns use every bank-year with reported loans, the loan-rate columns every bank-year with a non-missing loan rate.

The point estimates are ordered as the model predicts: intermediate-capital banks had the largest lending increase and the largest loan-rate decline. With only 22 low-capital and 37 high-capital banks, differences across groups are not precisely estimated: the lending differences are -0.249 (0.244) for Low minus Intermediate and -0.041 (0.118) for High minus Intermediate.

Risk Heterogeneity. A further implication of the model is that within the intermediate capital group, banks operating in higher risk environments should exhibit increases in lending larger than those experienced by banks operating in lower risk environments, because banks become effectively less risk-averse, allocating a larger share of their portfolios to risky loans. Because the risk-free rate is common across banks, a bank’s loan rate is a direct indicator of

the riskiness of the market in which it operates. To assess this implication qualitatively, we therefore classify the 431 intermediate capital group banks by their 2011 loan rate, relative to the quartiles of the pooled 2005–2017 loan-rate distribution: “low risk” (2011 loan rate below the 25th percentile), “medium risk” (between the 25th and the 75th percentile), and “high risk” (above the 75th percentile).

Table 3 reports the comparisons of average lending and loan rates pre- and post-announcement across these groups, which are broadly consistent with the implication of the model, as the increase in lending is more pronounced for banks operating in higher risk environments.

Table 3: Average Lending and Loan Rates of Intermediate Capital Group Banks by Risk

(1)		(2)	(3)	(4)	(5)	(6)	(7)
		2005–2011		2012–2017		Δ	$\Delta\%$
Risk groups	Banks	(log) loans	loan rate	(log) loans	loan rate	(log) loans	loan rate
Low risk	41	15.49 (0.31)	5.27 (0.13)	15.90 (0.25)	3.96 (0.11)	0.41 (0.15)	–24.81
Medium risk	339	13.57 (0.08)	6.21 (0.02)	14.22 (0.08)	4.62 (0.03)	0.65 (0.04)	–25.52
High risk	51	13.59 (0.25)	7.21 (0.15)	14.44 (0.31)	5.91 (0.19)	0.86 (0.16)	–18.05

Note. The sample is the 431 intermediate capital group banks of Table 2. Banks are assigned to a risk group on the basis of their 2011 loan rate relative to the 25th and 75th percentiles of the pooled 2005–2017 loan-rate distribution, and retain that assignment throughout; the group sizes therefore need not correspond to a quartile split of the 431 banks. Columns (2)–(7) are defined as in Table 2. Standard errors are clustered by bank. For Low, Medium, and High, the loan-rate level-change standard errors are 0.131, 0.026, and 0.142; the High-minus-Low difference in the change in (log) loans is 0.448 (0.212); a wild-cluster bootstrap of this difference with 9,999 replications and Webb weights gives $p = 0.035$, the same as the analytic cluster-robust p -value.

Within the intermediate-capital group, the lending increase rises monotonically with risk;²⁵ the difference between the high-risk and the low-risk group is 0.448 log points (SE 0.212), significant at the 5 percent level. The loan-rate decline is largest in the medium-risk group.

²⁵Because larger banks tend to charge lower loan rates, this classification is correlated with bank size (the correlation between the 2011 loan rate and (log) loans within the intermediate group is -0.26). Restricting attention to banks above the median size, for which the three groups are more evenly populated, the lending gradient is preserved: the increase in (log) loans equals 0.32, 0.42, and 0.61 across the low-, medium-, and high-risk groups, respectively.

Summary. Taken together, lending rose and loan rates fell in every capital and risk group, and all within-group changes were precisely estimated except lending among low-capital banks. The intermediate-capital group had the largest lending increase and loan-rate decline, as predicted by the model, although these differences relative to the small tail groups are imprecisely estimated. Within the intermediate-capital group, lending growth increased monotonically with risk. These patterns mirror the model’s distinction between quantity and price margins of adjustment in the intermediate region of bank capitalization.

6 The Optimal Length of the Transition Phase

We now ask how the total welfare varies with the length of the transition phase. We have seen in Section 5.1 that lending as a function of bank capital, the volatility of capital, and the distribution of losses, all change in a nontrivial way in response to a regulatory announcement. To aggregate this into a measure for total welfare, we have to take into account the distribution of capital during the transition phase.

Capital Accumulation in the Transition Phase. Figure 6 illustrates example paths of bank capital for different lengths of the transition phase. The paths are simulated under the assumption that the regulatory change is announced in $t = 0$, and phased in over a transition phase of one year (blue line) and three years (red dashed line), respectively. The black dashed-dotted line represents the status quo without a regulatory change.

Recall that the anticipation of tighter regulation increases the value of bank capital already in the transition phase. Hence, the payout boundary $\bar{E}_t$ increases by a discrete amount upon regulatory announcement and then converges to $\bar{E}_{new}$ by the end of the transition phase. Clearly, with a short transition phase (blue lines), the initial increase and the rate of convergence are much higher than with a long transition phase (red dashed lines). However, this channel is more likely to be relevant if banks are already well capitalized when the regulatory change is announced, such as in the example considered in the lower panel of Figure 6, where $E_0 = \bar{E}_{old}$.

The issuance boundary $\underline{E}_t$, on the contrary, remains close to $\underline{E}_{old}$ for most of the transition phase, rises only shortly before its end, and jumps to $\underline{E}_{new}$ when the tighter requirement takes effect (upper panel of Figure 6, where $E_0 = \underline{E}_{old}$). Banks thus issue little additional

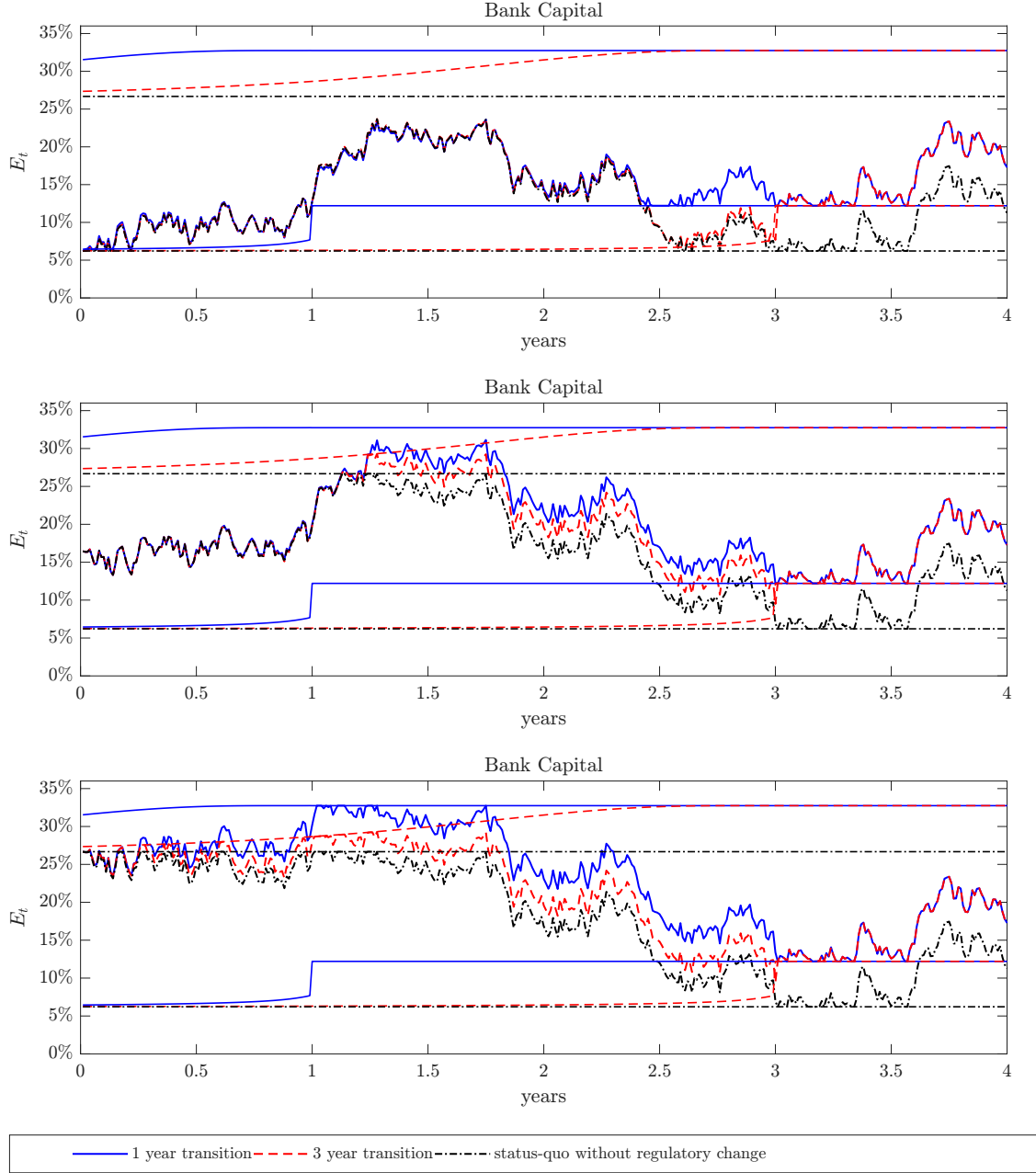


Figure 6: Illustrates sample paths for aggregate capital, E_t , as well as the evolution of the payout and issuance boundary, $\bar{E}$ and $\underline{E}$. We consider the introduction of the regulatory change from $\Lambda_{old} = 25$ to $\Lambda_{new} = 12.5$ over a one-year and a three-year transition phase. Initial capital is $E_0 = \underline{E}_{old}$ in the upper panel, at an intermediate level in the middle panel, and $E_0 = \bar{E}_{old}$ in the lower panel. Aggregate capital is expressed relative to the level of aggregate lending in the friction-less benchmark. Parameter values are as in Table 1.

capital in anticipation of the tightening, and poorly capitalized banks recapitalize to $\underline{E}_{new}$ only at the switch.

Welfare Analysis. Total welfare at a given point in time consists of the output of the productive sector Y , net of the social costs in case of financial crises, S , which are given by (7). The output of the productive sector is given by

$$Y_t := Y^l(R_t^l) + Y^h(R_t^h), \quad (30)$$

where output in market $i \in \{l, h\}$ is given by:²⁶

$$Y^i(R^i) = \frac{(\alpha^i + \beta^i R^i)(\alpha^i - \beta^i R^i)}{2\beta^i}. \quad (31)$$

Hence, all elements of total welfare are deterministic functions of aggregate capital (and the Poisson process governing the occurrence of systemic crises). To assess the welfare implications of a longer transition phase, we consider the present value of each of the two welfare components conditional on the initial level of aggregate capital E_0 . Output accrues as a flow, whereas the social cost S is incurred at the crisis times $(t_k)_{k \geq 1}$ of the Poisson process N , so that

$$\begin{aligned} PV(Y) &= \mathbf{E} \left[\int_0^\infty e^{-\rho t} Y_t dt \middle| E_0 \right], \\ PV(S) &= \mathbf{E} \left[\sum_{k \geq 1} e^{-\rho t_k} S_{t_k} \middle| E_0 \right] = \mathbf{E} \left[\int_0^\infty e^{-\rho t} \zeta S_t dt \middle| E_0 \right], \end{aligned} \quad (32)$$

where the second expression for $PV(S)$ replaces the jump process by its compensator and is the one we compute; accordingly, θ is the social cost per unit of the expected bailout shortfall. We computed the expectations by Monte Carlo simulation with 72,000 paths over a period of 100 years; Appendix C describes the design and reports Monte Carlo standard

²⁶See Appendix B for a derivation of aggregate output Y^i .

errors. Total welfare is then computed by adding up the two present values:²⁷

$$W := PV(Y) - PV(S). \quad (33)$$

The criterion W is a reduced-form objective for the regulator rather than a welfare function derived from household preferences. It counts the output financed by bank lending and the social cost of the public backstop, and it omits the transaction services of deposits.²⁸

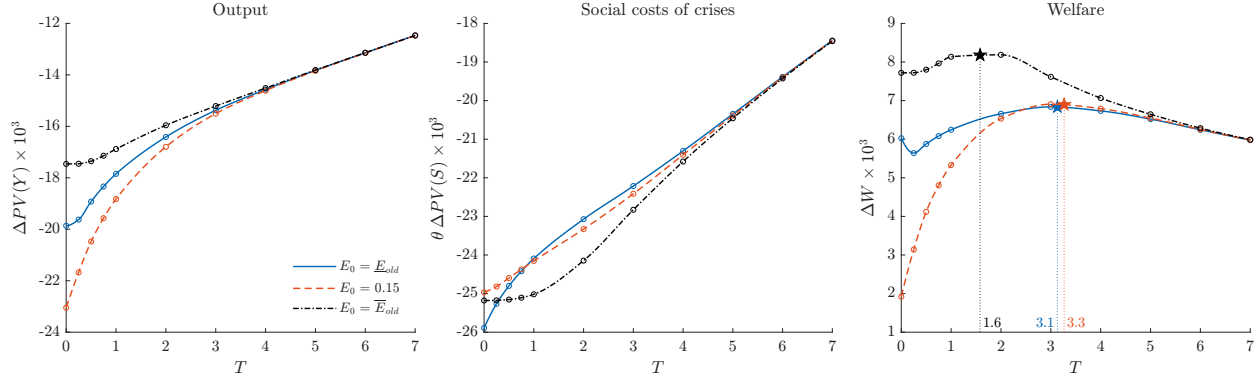


Figure 7: Welfare and its components as functions of the length of the transition phase T for a regulatory change from $\Lambda_{old} = 25$ to $\Lambda_{new} = 12.5$, announced in $t = 0$, at three levels of initial bank capital, $E_0 = \underline{E}_{old}$, $E_0 = 0.15$, and $E_0 = \bar{E}_{old}$, for the baseline $\theta = 0.50$. The left panel shows the present value of aggregate output, the middle panel the present value of the social costs of crises, and the right panel welfare, each relative to the status quo without a regulatory change; $T = 0$ is an immediate implementation. Markers are the simulated values, lines shape-preserving interpolations, and stars the welfare-maximizing transition length T^* , whose value is given at the dotted vertical lines. Parameter values are as in Table 1.

Costs and Benefits of a Longer Transition. Figure 7 plots welfare and its components as functions of the length of the transition phase T , relative to the status quo, for three levels

²⁷While a full moment-matched calibration of the equilibrium is beyond the scope of this paper, our numerical analysis offers some qualitative properties and comparative statics of the optimal length of the transition phase.

²⁸Deposits are supplied elastically at the fixed rate ρ and carry no liquidity premium in our model, so that their volume has no welfare value beyond the lending they finance. With a liquidity premium, as in Van den Heuvel (2008), a tighter requirement would carry an additional cost that W does not capture. We have also solved the model with a liquidity premium on deposits, whose value then enters W as a third component. The tightening forgoes part of this value, so that the welfare gain is slightly smaller, but the equilibrium and the optimal length of the transition, with and without announcement, are virtually unchanged.

of initial capital. The left panel shows the output cost of tighter regulation: the present value of output lies below the status quo, but the loss shrinks with the length of the transition, because banks have more time to build capital before the new requirement takes effect and, anticipating it, retain more of their earnings. The middle panel shows the other side: the tightening reduces the present value of social costs, but by less when it is introduced more slowly, since the new requirement then takes effect later. By (7), a unit of capital lowers the shortfall in a crisis one for one and a unit of lending raises it only by ϕ , so that this reduction stems mainly from the higher capitalization of banks under the new requirement and hardly from lower lending. For short transitions, the shape of the curves in both panels depends on initial capitalization, which determines how banks adjust under an immediate implementation.

At intermediate capitalization, $E_0 = 0.15$, output is concave in T : the gain from delay is largest in the first months and declines thereafter, since an immediate implementation there causes the most severe credit crunch (recall from Figure 2 that in this region the requirement is slack under Λ_{old} and binds under Λ_{new}). Social costs, in contrast, are close to linear in T . Banks do not recapitalize at this level of capital, and the cut in lending lowers the shortfall only little, so that social costs fall only as banks build up capital under the new requirement. A delay postpones this gradual build-up: capital stays close to its status-quo path until it reaches one of the boundaries, and from an intermediate level it is unlikely to do so during a short transition (middle panel of Figure 6). Each month of delay therefore costs about the same.

At $E_0 = \underline{E}_{old}$ the adjustment is the reverse: an immediate implementation makes banks recapitalize to the new issuance boundary $\underline{E}_{new}$ at once, but hardly changes lending, which is constrained under both requirements ($\Lambda_{old}\underline{E}_{old} \approx \Lambda_{new}\underline{E}_{new}$). For output, a very short transition changes little: it only delays this recapitalization to the date at which the issuance boundary jumps (upper panel of Figure 6), and with it the adjustment that follows, so that the output gained during the transition is lost after it. Output is therefore convex in T for delays of up to about half a year and concave beyond. For social costs, the first months of delay are instead the most costly, because they forgo the reduction in the shortfall that the immediate recapitalization brings from the first day; later months cost less because banks build capital during the transition. Social costs are thus concave in T . Since the first months of delay cost more in stability than they gain in output, welfare at this level of capital first

falls in T , and an immediate implementation is a second, local maximum.

At $E_0 = \bar{E}_{old}$, where neither requirement binds, banks respond to the announcement itself. They price the tightening in almost at once (Figure 4): lending is close to its new-regime level throughout a short transition and stays near its status-quo level only in months well ahead of the switch. Delay therefore adds little output until the transition is long enough to contain such months, and output is convex in T for delays of up to about one year and concave beyond. Banks also retain their earnings from the moment of the announcement, since $\bar{E}_t > \bar{E}_{old}$ throughout the transition (lower panel of Figure 6), so a short delay costs almost no stability. The cost of delay rises only for longer transitions, over which the payout boundary starts closer to $\bar{E}_{old}$ and capital accumulates more slowly, so that social costs are convex in T .

Beyond about five years the three curves coincide in each panel: the distribution of capital is then close to its stationary distribution under the old regime well before the switch, whatever its initial level, and a further delay only postpones the same difference between the two regimes. Since the tightening raises welfare, such a postponement lowers it: output then rises more slowly in T than social costs. The welfare-maximizing transition length, marked in the right panel, equates the marginal gain in output from a longer transition with its marginal social cost. At $E_0 = \underline{E}_{old}$ and at $E_0 = 0.15$, where output is concave in T beyond the first months, this is the case after about three years. At $E_0 = \bar{E}_{old}$, where the marginal gain in output is smaller and social costs are convex in T , this is the case after about one and a half years.

At the optimal transition length, the tightening raises welfare by between 2.1 and 2.5 percent of one year's output in the status quo. (One year's output in the status quo equals 0.32, so that one percent corresponds to 3.2 on the scale of Figure 7.)

The Importance of the Anticipation Effect. To isolate the anticipation effect, we compare the optimal transition with a benchmark in which the regulator decides in $t = 0$ that the tightening takes effect after T years but does not announce it. Until T , lending and capital accumulation then follow the status quo, and a delay only postpones the switch between the two regimes: the benchmark objective is the status-quo welfare flow up to T plus the discounted value of switching at the level of capital reached by then. Figure 8 compares

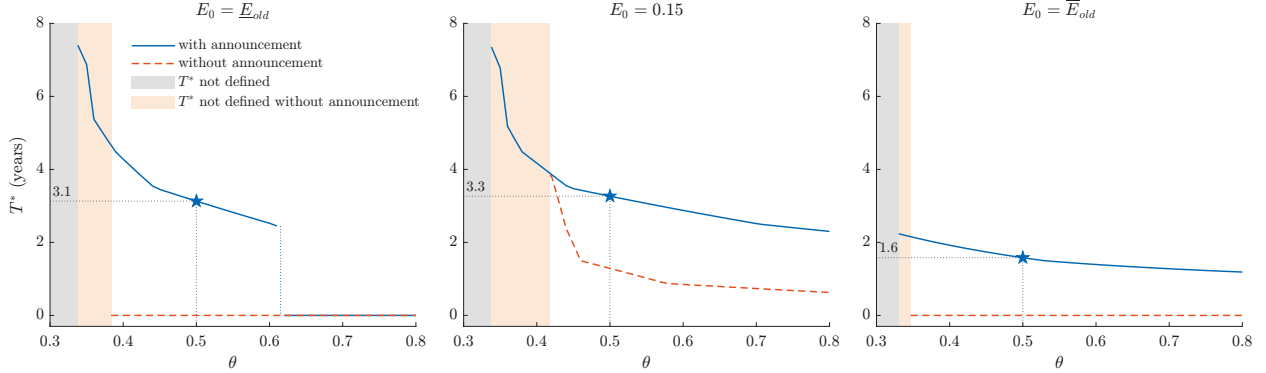


Figure 8: Welfare-maximizing transition phase T^* as a function of the social-cost weight θ , with announcement (solid line) and in the benchmark without announcement (dashed line), for $E_0 = \underline{E}_{old}$ (left), $E_0 = 0.15$ (middle), and $E_0 = \overline{E}_{old}$ (right). T^* is not defined where no transition length raises welfare above the status quo: in the grey region in both cases, and in the light orange region without announcement. The dotted vertical line marks the baseline $\theta = 0.50$, the star and the dotted horizontal line the corresponding T^* with announcement (the optimum marked in Figure 7). The dotted vertical segment in the left panel marks a jump of T^* , where the welfare maximum switches between two local maxima. Parameter values are as in Table 1.

the two optimal transition lengths across θ for the three levels of E_0 of Figure 7.²⁹

With the announcement, the optimal transition length is interior and declines in θ : the more weight the regulator puts on the social costs of crises, the sooner the tighter requirement should take effect. In the left panel it drops to an immediate implementation for θ of about 0.62 and above. Without the announcement, the optimum is an immediate implementation at low and high capitalization, and at intermediate capitalization it is interior, but shorter than the optimal transition with the announcement. Where both are defined, the benchmark transition is thus never the longer one. At the baseline ($\theta = 0.50$), it is shorter by 3.1, 2.0, and 1.6 years in the three panels. The difference between welfare under the optimal announced transition and under the best transition without announcement measures the contribution of the anticipation effect: it amounts to 12, 46, and 6 percent of the welfare gain from the tightening at the three levels of initial capital, the entire value of the transition at low and high capitalization and about two thirds of it at intermediate capitalization (Table 4,

²⁹ T^* is defined only where the tightening raises welfare above the status quo for some transition length. Since the welfare gain from the tightening increases in θ for every transition length, this is the case for θ above a threshold, which is about 0.34 with the announcement (0.33 at $E_0 = \overline{E}_{old}$) and 0.38, 0.42, and 0.35 without it at $E_0 = \underline{E}_{old}$, $E_0 = 0.15$, and $E_0 = \overline{E}_{old}$. The shaded regions in Figure 8 mark the values of θ below these thresholds.

columns (4) and (5)). Without an announcement, a delay only moves the distribution of capital towards its stationary distribution under the old requirement, which it has largely reached after two years; with the announcement, the transition itself builds capital, because the higher payout boundary makes banks retain their earnings. A regulator who did not account for the anticipation effect would therefore choose a transition that is too short. To measure the cost, we evaluate welfare with the anticipation effect at the transition length that is optimal in the benchmark: the regulator announces and implements the transition it would choose if banks did not respond to the announcement. Relative to the optimal transition, 12, 16, and 6 percent of the potential welfare gain from the tighter requirement are then lost at low, intermediate, and high capitalization. At low and high capitalization, where this regulator implements at once, this is the entire part of the gain that depends on the timing. At levels of initial capital between those shown, the loss reaches about two fifths of the gain (39 percent at $E_0 = 0.085$, where the benchmark still calls for an immediate implementation; see Table 4).

7 Conclusion

Regulators are often criticized for being slow in implementing post-crisis regulatory reforms. Basel III was announced in 2010 with a transition period that phased in the higher capital requirements between 2013 and 2019, and its final standards, agreed in 2017, were to take effect in 2022 and to be phased in over a further five years (Basel Committee on Banking Supervision, 2011; 2017). To our knowledge, this paper is the first to characterize the welfare-optimal length of a phase-in when banks' payout and issuance policies respond to the announcement of the reform, and it shows the virtue of being slow.

To this end, we develop a stylized dynamic equilibrium model in which banks finance risky loans to the real sector by equity and liquid deposits. Banks incur flotation costs when issuing equity and thus build up capital buffers from retained earnings. Furthermore, despite being perfectly competitive, banks require a strictly positive lending premium because of the following equilibrium mechanism. If banks retain earnings, aggregate capital increases, which leads to a larger aggregate loan supply and, thus, to a lower equilibrium loan rate. With lower profits from lending, the market-to-book ratio of any bank's equity decreases. Hence, banks become effectively risk-averse, as their shareholder value resembles a concave

utility function. This yields three key testable implications of our model: aggregate lending increases in aggregate capital, while the equilibrium loan rates and the market-to-book ratio of equity decrease in aggregate capital.

After the implementation of a regulatory reform, the tighter capital requirements restrict aggregate loan supply more severely. This increases the profitability of banks' loan making business and raises the market-to-book ratio of equity. The announcement of such a regulatory tightening changes banks' financing policies: Anticipating a higher market-to-book ratio, banks retain more of their earnings during the transition phase, which has the benefit that capital is accumulated more quickly: better capitalized banks lend more and leave a smaller shortfall in a crisis, and the tighter regulation is less likely to be binding when it is implemented. The timing of a reform cannot be assessed without this response. A regulator who does not account for it sees a delay as merely shifting the switch between the two regimes in time, chooses a transition that is too short, and forfeits up to two fifths of the welfare gain from the tightening. The anticipation effect itself accounts for up to almost half of this gain. The timing matters most at intermediate levels of bank capital, where an immediate implementation would forfeit most of the gain, and not, as one might expect, where banks are least capitalized.

The anticipation effect also changes lending during the transition phase at a given level of capital, and not in the same direction everywhere. When banks are poorly capitalized, lending is determined by the binding capital requirement and does not respond to the announcement. When they are very well capitalized, the future scarcity of loans is fully "priced-in": loan rates rise and lending falls towards its level under the new requirement. In between, where the requirement becomes slack, the market-to-book ratio of equity falls more slowly in aggregate capital, so that banks are effectively less risk averse, loan rates fall, and lending rises above its level without the reform. The announcement of a regulatory tightening can thus raise aggregate lending. This is consistent with the evidence that aggregate lending grew after the jurisdictional announcement dates of the Basel III reforms (Basel Committee on Banking Supervision, 2022, Section 8.1.3 and Annex A.18).

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References

- Admati, A. R., DeMarzo, P. M., Hellwig, M. F., & Pfleiderer, P. C. (2013). *Fallacies, Irrelevant Facts, and Myths in the Discussion of Capital Regulation: Why Bank Equity is Not Socially Expensive* (Working Paper). Stanford University.
- Adrian, T., & Boyarchenko, N. (2015). *Intermediary Leverage Cycles and Financial Stability* (Staff Report No. 567). Federal Reserve Bank of New York.
- Aiyar, S., Calomiris, C. W., & Wieladek, T. (2016). How does credit supply respond to monetary policy and bank minimum capital requirements? *European Economic Review*, 82, 142–165.

- Allen, F., Carletti, E., & Marquez, R. (2011). Credit Market Competition and Capital Regulation. *The Review of Financial Studies*, 24(4), 983–1018.
- Asquith, P., & Mullins, D. W. (1986). Equity issues and offering dilution. *Journal of Financial Economics*, 15(1–2), 61–89.
- Asvanunt, A., & Richardson, S. (2017). The Credit Risk Premium. *The Journal of Fixed Income*, 26(3), 6–24.
- Ballard, C. L., Shoven, J. B., & Whalley, J. (1985). General Equilibrium Computations of the Marginal Welfare Costs of Taxes in the United States. *American Economic Review*, 75(1), 128–138.
- Basel Committee on Banking Supervision. (2006). *International convergence of capital measurement and capital standards: A revised framework – comprehensive version* (BCBS Standards). Bank for International Settlements.
- Basel Committee on Banking Supervision. (2011). *Basel III: A global regulatory framework for more resilient banks and banking systems* (BCBS Standards). Bank for International Settlements.
- Basel Committee on Banking Supervision. (2017). *Basel III: Finalising post-crisis reforms* (BCBS Standards). Bank for International Settlements.
- Basel Committee on Banking Supervision. (2019). *The costs and benefits of bank capital – a review of the literature* (BCBS Working Paper No. 37). Bank for International Settlements.
- Basel Committee on Banking Supervision. (2022). *Evaluation of the impact and efficacy of the Basel III reforms* (BCBS Evaluation Report). Bank for International Settlements.
- Begenau, J. (2020). Capital requirements, risk choice, and liquidity provision in a business-cycle model. *Journal of Financial Economics*, 136(2), 355–378.
- Begenau, J., & Landvoigt, T. (2022). Financial Regulation in a Quantitative Model of the Modern Banking System. *The Review of Economic Studies*, 89(4), 1748–1784.
- Behn, M., Haselmann, R., & Wachtel, P. (2016). Procyclical Capital Regulation and Lending. *The Journal of Finance*, 71(2), 919–956.

- Black, L., Floros, I. V., & Sengupta, R. (2016). *Raising Capital When the Going Gets Tough: U.S. Bank Equity Issuance from 2001 to 2014* (Working Paper No. 16-05). Federal Reserve Bank of Kansas City.
- Bolton, P., Chen, H., & Wang, N. (2011). A Unified Theory of Tobin’s q , Corporate Investment, Financing, and Risk Management. *The Journal of Finance*, 66(5), 1545–1578.
- Bolton, P., Chen, H., & Wang, N. (2013). Market timing, investment, and risk management. *Journal of Financial Economics*, 109(1), 40–62.
- Bolton, P., Li, Y., Wang, N., & Yang, J. (2025). Dynamic Banking and the Value of Deposits. *The Journal of Finance*, 80(4), 2063–2105.
- Boyson, N. M., Fahlenbrach, R., & Stulz, R. M. (2016). Why Don’t All Banks Practice Regulatory Arbitrage? Evidence from Usage of Trust-Preferred Securities. *Review of Financial Studies*, 29(7), 1821–1859.
- Brunnermeier, M. K., & Sannikov, Y. (2014). A Macroeconomic Model with a Financial Sector. *American Economic Review*, 104(2), 379–421.
- Cerra, V., & Saxena, S. C. (2008). Growth Dynamics: The Myth of Economic Recovery. *American Economic Review*, 98(1), 439–457.
- Clerc, L., Derviz, A., Mendicino, C., Moyen, S., Nikolov, K., Stracca, L., Suarez, J., & Vardoulakis, A. P. (2015). Capital regulation in a macroeconomic model with three layers of default. *International Journal of Central Banking*, 11(3), 9–63.
- Corbae, D., & D’Erasmus, P. (2021). Capital Buffers in a Quantitative Model of Banking Industry Dynamics. *Econometrica*, 89(6), 2975–3023.
- Crawford, G. S., Pavanini, N., & Schivardi, F. (2018). Asymmetric Information and Imperfect Competition in Lending Markets. *American Economic Review*, 108(7), 1659–1701.
- Davison, L. (2020). *Two Crises: A Comparison* (Staff Studies Report No. 2020-02). FDIC.
- Davydiuk, T. (2017). *Dynamic Bank Capital Requirements* (Working Paper). Carnegie Mellon University.

- De Jonghe, O., Dewachter, H., & Ongena, S. (2020). Bank capital (requirements) and credit supply: Evidence from pillar 2 decisions. *Journal of Corporate Finance*, 60, 101518.
- De Nicolò, G., Gamba, A., & Lucchetta, M. (2014). Microprudential Regulation in a Dynamic Model of Banking. *The Review of Financial Studies*, 27(7), 2097–2138.
- DeAngelo, H., & Stulz, R. M. (2015). Liquid-claim production, risk management, and bank capital structure: Why high leverage is optimal for banks. *Journal of Financial Economics*, 116(2), 219–236.
- Décamps, J.-P., Mariotti, T., Rochet, J.-C., & Villeneuve, S. (2011). Free Cash Flow, Issuance Costs, and Stock Prices. *The Journal of Finance*, 66(5), 1501–1544.
- DeFusco, A. A., & Paciorek, A. (2017). The Interest Rate Elasticity of Mortgage Demand: Evidence from Bunching at the Conforming Loan Limit. *American Economic Journal: Economic Policy*, 9(1), 210–240.
- Diamond, D. W., & Rajan, R. G. (2000). A Theory of Bank Capital. *The Journal of Finance*, 55(6), 2431–2465.
- Elenev, V., Landvoigt, T., & Van Nieuwerburgh, S. (2021). A Macroeconomic Model with Financially Constrained Producers and Intermediaries. *Econometrica*, 89(3), 1361–1418.
- Fonseca, A. R., & González, F. (2010). How bank capital buffers vary across countries: The influence of cost of deposits, market power and bank regulation. *Journal of Banking & Finance*, 34(4), 892–902.
- Fraisse, H., Lé, M., & Thesmar, D. (2020). The Real Effects of Bank Capital Requirements. *Management Science*, 66(1), 5–23.
- Freixas, X., & Rochet, J.-C. (2023). *Microeconomics of Banking* (3rd ed.). MIT Press.
- Gertler, M., & Karadi, P. (2011). A model of unconventional monetary policy. *Journal of Monetary Economics*, 58(1), 17–34.
- Gorton, G., & Winton, A. (2017). Liquidity Provision, Bank Capital, and the Macroeconomy. *Journal of Money, Credit and Banking*, 49(1), 5–37.

- Gropp, R., Mosk, T., Ongena, S., & Wix, C. (2019). Banks Response to Higher Capital Requirements: Evidence from a Quasi-Natural Experiment. *The Review of Financial Studies*, 32(1), 266–299.
- He, Z., & Krishnamurthy, A. (2012). A Model of Capital and Crises. *The Review of Economic Studies*, 79(2), 735–777.
- Hellmann, T. F., Murdock, K. C., & Stiglitz, J. E. (2000). Liberalization, Moral Hazard in Banking, and Prudential Regulation: Are Capital Requirements Enough? *American Economic Review*, 90(1), 147–165.
- Hoffmann, F., & Pfeil, S. (2021). *Dividend Regulation and Risk-Taking in a Dynamic Model of Banking* (Working Paper). University of Groningen.
- Homar, T., & van Wijnbergen, S. J. G. (2017). Bank recapitalization and economic recovery after financial crises. *Journal of Financial Intermediation*, 32, 16–28.
- Hugonnier, J., & Morellec, E. (2017). Bank capital, liquid reserves, and insolvency risk. *Journal of Financial Economics*, 125(2), 266–285.
- Jordà, Ó., Richter, B., Schularick, M., & Taylor, A. M. (2021). Bank Capital Redux: Solvency, Liquidity, and Crisis. *The Review of Economic Studies*, 88(1), 260–286.
- Klimenko, N., Pfeil, S., & Rochet, J.-C. (2017). A simple macroeconomic model with extreme financial frictions. *Journal of Mathematical Economics*, 68, 92–102.
- Kondor, P., & Vayanos, D. (2019). Liquidity Risk and the Dynamics of Arbitrage Capital. *The Journal of Finance*, 74(3), 1139–1173.
- Laeven, L., & Valencia, F. (2013). The Real Effects of Financial Sector Interventions during Crises. *Journal of Money, Credit and Banking*, 45(1), 147–177.
- Laeven, L., & Valencia, F. (2020). Systemic Banking Crises Database II. *IMF Economic Review*, 68(2), 307–361.
- Malherbe, F. (2020). Optimal Capital Requirements over the Business and Financial Cycles. *American Economic Journal: Macroeconomics*, 12(3), 139–174.
- Martinez-Miera, D., & Suarez, J. (2014). *Banks’ endogenous systemic risk taking* (Working Paper). CEMFI.

- Mendicino, C., Nikolov, K., Suarez, J., & Supera, D. (2020). Bank capital in the short and in the long run. *Journal of Monetary Economics*, 115, 64–79.
- Mésonnier, J.-S., & Monks, A. (2015). Did the EBA Capital Exercise Cause a Credit Crunch in the Euro Area? *International Journal of Central Banking*, 11(3), 75–117.
- Miles, D., Yang, J., & Marcheggiano, G. (2013). Optimal Bank Capital. *The Economic Journal*, 123(567), 1–37.
- Milne, A., & Whalley, A. E. (2001). *Bank Capital and Incentives for Risk-Taking* (Finance Group Research Paper No. 15). Warwick Business School.
- Morrison, A. D., & White, L. (2005). Crises and Capital Requirements in Banking. *American Economic Review*, 95(5), 1548–1572.
- Phelan, G. (2016). Financial Intermediation, Leverage, and Macroeconomic Instability. *American Economic Journal: Macroeconomics*, 8(4), 199–224.
- Repullo, R. (2004). Capital requirements, market power, and risk-taking in banking. *Journal of Financial Intermediation*, 13(2), 156–182.
- Repullo, R., & Suarez, J. (2013). The Procyclical Effects of Bank Capital Regulation. *The Review of Financial Studies*, 26(2), 452–490.
- Schliephake, E. (2016). Capital Regulation and Competition as a Moderator for Banking Stability. *Journal of Money, Credit and Banking*, 48(8), 1787–1814.
- Sundaresan, S., & Wang, Z. (2023). Strategic Bank Liability Structure Under Capital Requirements. *Management Science*, 69(10), 6349–6368.
- Van den Heuvel, S. J. (2008). The welfare cost of bank capital requirements. *Journal of Monetary Economics*, 55(2), 298–320.
- Wang, Y., Whited, T. M., Wu, Y., & Xiao, K. (2022). Bank Market Power and Monetary Policy Transmission: Evidence from a Structural Estimation. *The Journal of Finance*, 77(4), 2093–2141.
- World Bank. (2020). *Global Financial Development Report 2019/2020: Bank Regulation and Supervision a Decade after the Global Financial Crisis*.

Appendix A Omitted Proofs

Proof of Proposition 1.

Note first that the equilibrium loan rates in both markets have to be at least as high as the common discount rate, i.e.,

$$R^i(E) \geq \rho, \quad (\text{A.1})$$

for all E . We denote the market-to-book ratio by $u_b(E)$ (and loan rates by $R_b^i(E)$) for all E where the capital requirement is binding and by $u_s(E)$ (and $R_s^i(E)$) for all E where it is slack. Lemma A.1 derives an auxiliary result for the proof of the uniqueness of E^* .

Lemma A.1. *Assume that the constraint is slack over a region $E \in [E^*, \bar{E}]$ with $E^* > \underline{E}$, then it holds for $E \in [E^*, \bar{E})$ that*

$$(i) \quad u'_s(E) < 0,$$

$$(ii) \quad u''_s(E) > 0.$$

Proof. Substituting boundary conditions (16) and (18) into HJB (22) implies that u''_s is positive at the top:

$$u''_s(\bar{E}) = \frac{2}{(\sigma^l)^2 + (\sigma^h)^2} \zeta > 0, \quad (\text{A.2})$$

where we used $R^i(\bar{E}) = \rho$ from (28) together with the normalization $L^i(\rho) = 1$, so that $K^i(\bar{E}) = 1$. By continuity, it must therefore hold that $u'_s(\bar{E} - \epsilon) < 0$ for a small ϵ . Now assume that $u'_s(E) \geq 0$ for some $E \in [E^*, \bar{E})$ and let $\hat{E} := \sup\{E < \bar{E} : u'_s(E) \geq 0\}$. By continuity we have $u'_s(\hat{E}) = 0$ and $u''_s(\hat{E}) \leq 0$, implying that $u_s(\hat{E}) = \frac{(\sigma^l K^l)^2 + (\sigma^h K^h)^2}{2\zeta} u''_s(\hat{E}) \leq 0$. This is a contradiction, since $u_s(\bar{E}) = 1$ and $u'_s(E) < 0$ for $E \in (\hat{E}, \bar{E})$. The first claim of Lemma A.1 therefore follows. The second claim follows immediately from (22) together with the first claim and (A.1). $\square$

We next characterize E^* and delimit the region of the state space in which the capital requirement can bind at all. By definition, E^* , the lowest point at which the constraint is not strictly binding, satisfies $L^l(R_s^l) + L^h(R_s^h) = \Lambda E$. From (25), this condition can be written as

$$\frac{(m^l - \rho)u_s(E)}{n^l u_s(E) - \sigma^{l2} u'_s(E)} + \frac{(m^h - \rho)u_s(E)}{n^h u_s(E) - \sigma^{h2} u'_s(E)} = \Lambda E, \quad (\text{A.3})$$

where we have rewritten (9) for convenience as

$$R^i = (L^i)^{-1} = m^i - n^i K^i$$

with $m^i := \alpha^i / \beta^i$ and $n^i := 1 / \beta^i$. The left-hand side of (A.3) is aggregate lending in the region where the requirement is slack. Since $R^i(E) \geq \rho$ by (A.1) and loan demand is strictly decreasing, aggregate lending never exceeds its frictionless level, $L^l(R^l(E)) + L^h(R^h(E)) \leq L^l(\rho) + L^h(\rho)$. The capital requirement can therefore bind only at capitalizations

$$E \leq \frac{L^l(\rho) + L^h(\rho)}{\Lambda}, \quad (\text{A.4})$$

which for the normalization $L^i(\rho) = 1$ equals $2/\Lambda$.

Both sides of (A.3) are strictly increasing in E : the right-hand side by inspection, and the left-hand side because loan rates are strictly decreasing in aggregate capital by Corollary 1, so that lending rises with E in both segments. The switch point E^* is thus located where the slack-region lending schedule meets the regulatory line ΛE , and the requirement binds on a single interval $[\underline{E}, E^*]$ whenever that crossing occurs from below, i.e. whenever slack-region lending rises less steeply in E than ΛE does. In every equilibrium computed for this paper this is the case by a wide margin: aggregate lending in the slack region rises at 2.1% (under

Λ_{old}) and 4.5% (under Λ_{new}) of the rate at which the leverage limit relaxes, and the two schedules cross exactly once on the admissible range (A.4). We also verify that the shadow costs are strictly positive inside the binding region: in both regimes and in both segments, $\eta^i(E)$ declines from about 0.06 at $\underline{E}$ to zero at E^* and is positive at every grid point in between, so that the requirement indeed binds on $[\underline{E}, E^*)$.

After having established the above regularities, we now show how to construct the equilibrium. To solve for the equilibrium functions $u(E)$ and $R^i(E)$, we first consider a candidate value for the issuance boundary, $\underline{E}_c$, and solve the system (22) – (24) by value function iteration, subject to boundary conditions (15) and (20), i.e.,

$$u_b(\underline{E}_c; \underline{E}_c) - (1 + \gamma) = \frac{\partial}{\partial E} u_b(\underline{E}_c; \underline{E}_c) = 0.$$

Here, we adopt the notation $u_b(E; \underline{E}_c)$ to emphasize that the market-to-book ratio is a function of E and parameterized by the candidate value $\underline{E}_c$, for which the remaining boundary conditions are not necessarily satisfied. We can then determine, also parameterized by $\underline{E}_c$, the critical level of aggregate equity,

$$E_c^* := E^*(\underline{E}_c),$$

at which the constraint imposed by (8) becomes slack, that is,

$$\frac{(m^l - \rho)u_b(E_c^*; \underline{E}_c)}{n^l u_b(E_c^*; \underline{E}_c) - \sigma^{l^2} u_b'(E_c^*; \underline{E}_c)} + \frac{(m^h - \rho)u_b(E_c^*; \underline{E}_c)}{n^h u_b(E_c^*; \underline{E}_c) - \sigma^{h^2} u_b'(E_c^*; \underline{E}_c)} = \Lambda E_c^*. \quad (\text{A.5})$$

Note that through the respective boundaries, also the equilibrium loan rates $R_b^i(E; \underline{E}_c)$ are parameterized by the candidate value $\underline{E}_c$.

Turning next to the region where the regulatory constraint is slack, we can determine the equilibrium loan rates $R_s^i(E; \underline{E}_c)$ by solving (27) subject to the following boundary condition

$$R_s(E_c^*; \underline{E}_c) = R_b(E_c^*; \underline{E}_c), \quad (\text{A.6})$$

which ensures continuity of the loan rate at the point E_c^* . By substituting either $R_s^l(E; \underline{E}_c)$ or $R_s^h(E; \underline{E}_c)$ into the respective first order condition for banks' leverage (25), we can compute the market-to-book ratio in the region where the constraint is slack:

$$u_s(E; \underline{E}_c) = u_s(E_c^*; \underline{E}_c) \times \exp\left(-\int_{E_c^*}^E \frac{R_s^i(q; \underline{E}_c) - \rho}{(\sigma^i)^2 L^i(R_s^i(q; \underline{E}_c))} dq\right), \quad (\text{A.7})$$

It is important to stress that $u_s(E; \underline{E}_c)$ is parameterized by the candidate $\underline{E}_c$ first, through the loan rates from (A.6) and, second, by imposing value-matching at E_c^* in (A.7), i.e.,

$$u_s(E_c^*; \underline{E}_c) = u_b(E_c^*; \underline{E}_c).$$

Next, we determine (also parameterized by the candidate $\underline{E}_c$) the payout boundary $\bar{E}(\underline{E}_c)$ by using the boundary condition (18):

$$\frac{\partial}{\partial E} u_s(\bar{E}(\underline{E}_c); \underline{E}_c) = 0.$$

Finally, note that we have constructed a continuous, piecewise function

$$u(E; \underline{E}_c) = \begin{cases} u_b(E; \underline{E}_c) & \text{if } E \leq E_c^*, \\ u_s(E; \underline{E}_c) & \text{if } E > E_c^*. \end{cases}$$

The same applies to $R^i(\cdot)$. Since all endogenous objects are parameterized by the candidate value $\underline{E}_c$, it remains to pin down $\underline{E}$ by the remaining boundary condition (16). Equivalently, $\underline{E}$ is a zero of the shooting residual

$$\Phi(\underline{E}_c) := u(\overline{E}(\underline{E}_c); \underline{E}_c) - 1, \tag{A.8}$$

which we solve numerically: in our computations, Φ is continuous and crosses zero exactly once over the admissible range of $\underline{E}_c$, locating a unique issuance boundary $\underline{E}$ to a tolerance of 10^{-8} . □

Proof of Corollary 1.

We show this for R^l ; R^h is analogous. From (25) we get

$$\frac{u'(E)}{u(E)} = -\frac{R^l(E) - \rho}{L^l(R^l(E))(\sigma^l)^2}. \quad (\text{A.9})$$

Differentiating (A.9) yields

$$\frac{u''(E)}{u(E)} = \frac{\left(\frac{R^l(E) - \rho}{\sigma^l}\right)^2 - \frac{\partial R^l(E)}{\partial E} \left(L^l(R^l(E)) - (R^l(E) - \rho) \frac{\partial L^l(R^l)}{\partial R^l}\right)}{\left(L^l(R^l(E))\sigma^l\right)^2}. \quad (\text{A.10})$$

Dividing HJB (22) by $u(E)$, substituting (A.9) and (A.10) and rearranging yields

$$\begin{aligned} \frac{\partial R^l(E)}{\partial E} = & -\psi(E) \frac{L^l(R^l(E))(R^l(E) - \rho)}{L^l(R^l(E)) - \frac{\partial L^l(R^l)}{\partial R^l}(R^l(E) - \rho)} \\ & + \frac{2 \left[\frac{R^l(E) - \rho}{L^l(R^l(E))(\sigma^l)^2} - \frac{R^h(E) - \rho}{L^h(R^h(E))(\sigma^h)^2} \right] (L^h(R^h(E))\sigma^h)^2 (R^l(E) - \rho) L^l(R^l(E))}{\left[\left(\sigma^l L^l(R^l(E))\right)^2 + \left(\sigma^h L^h(R^h(E))\right)^2 \right] \left[L^l(R^l(E)) - \frac{\partial L^l(R^l)}{\partial R^l}(R^l(E) - \rho) \right]}, \end{aligned} \quad (\text{A.11})$$

where

$$\begin{aligned} \psi(E) := & \frac{R^l(E) - \rho}{L^l(R^l(E))(\sigma^l)^2} + \frac{2E\rho + 2\zeta L^l(R^l(E))(\sigma^l)^2 / (R^l(E) - \rho)}{\left(\sigma^l L^l(R^l(E))\right)^2 + \left(\sigma^h L^h(R^h(E))\right)^2} \\ = & \frac{2E\rho + L^l(R^l(E))(R^l(E) - \rho) + L^h(R^h(E))(R^h(E) - \rho)}{\left(\sigma^l L^l(R^l(E))\right)^2 + \left(\sigma^h L^h(R^h(E))\right)^2} \\ & + \frac{2\zeta}{L^l(R^l(E))(R^l(E) - \rho) + L^h(R^h(E))(R^h(E) - \rho)}, \end{aligned} \quad (\text{A.12})$$

and the second equality uses (25). Condition (25) implies that the term in square brackets in (A.11) equals zero and thus (27) follows. Equation (28) follows from (25) and boundary conditions (16) and (18). Since the market-to-book ratio is strictly decreasing for $E < \bar{E}$ (as we show in the Proof of Proposition 1), condition (25) implies $R^i(E) > \rho$ for $E \in [E^*, \bar{E})$ and thus $\psi(E) > 0$. Since aggregate loan demand is strictly decreasing, i.e., $\partial L^i(R^i)/\partial R^i < 0$ for $R^i \in [\rho, \hat{R}^i]$, the denominator of (27) is positive as well and thus $\partial R^i(E)/\partial E < 0$ for $E \in [E^*, \bar{E})$. At the payout boundary, where $R^i(\bar{E}) = \rho$, equation (27) holds in the limiting sense: evaluating (22) at $\bar{E}$, where $u'(\bar{E}) = 0$ and $u(\bar{E}) = 1$ by (18) and (16), yields $u''(\bar{E}) = 2\zeta/[(\sigma^l)^2 + (\sigma^h)^2]$ (using $L^i(\rho) = 1$), and differentiating (25) at $\bar{E}$ then gives $\lim_{E \uparrow \bar{E}} \partial R^i(E)/\partial E = -2\zeta(\sigma^i)^2/[(\sigma^l)^2 + (\sigma^h)^2] < 0$.

Appendix B Aggregate Loan Demand and Output

The aggregate loan demand in (9) can be microfounded by the assumption that the productive sector in each of the two markets $i \in \{l, h\}$ is populated by a continuum of firms that differ with respect to their productivity x . That is, at any time t , a firm with productivity x can convert one unit of the consumption good into $1 + xdt$ units of the consumption good at time $t + dt$. Firms' productivity is uniformly distributed over the interval $[\rho, \frac{\alpha^i}{\beta^i}]$,³⁰ i.e., the density function of x equals

$$f^i(x) = \beta^i \tag{B.1}$$

³⁰The uniform distribution is adopted for tractability: it produces the linear loan-demand specification $L^i(R^i) = \alpha^i - \beta^i R^i$ used throughout the paper. Empirical firm-productivity distributions are typically right-skewed (log-normal or Pareto-tail), which would generate a convex loan-demand curve. The qualitative results of this paper (monotonicity of $R^i(E)$ in E , the shape of the binding/slack regions, the announcement-induced flattening of $u(E, t)$, and the welfare ranking across transition lengths) rely on the demand curve being decreasing and twice-differentiable, both of which hold under the right-skewed alternatives. We adopt the uniform-linear pair to keep the equilibrium characterization in closed form.

for $\rho \leq x \leq \frac{\alpha^i}{\beta^i}$ and zero otherwise. The rate of return on investment for a firm with productivity x is always equal to its productivity minus the rental rate of capital. Hence, a firm takes up a bank loan and invests if and only if its productivity exceeds the prevailing rental rate of capital, i.e., if $x > R^i$. Hence, aggregate demand for bank loans equals

$$\begin{aligned} L^i(R^i) &= \int_{R^i}^{\frac{\alpha^i}{\beta^i}} f^i(x) dx \\ &= \alpha^i - \beta^i R^i, \end{aligned} \tag{B.2}$$

which is equation (9). We allow for $\beta^h \leq \beta^l$, so that high-risk loans may be more profitable than low-risk loans. Furthermore, we normalize to one the loan demands in the first-best benchmark, $L^i(\rho) = 1$. Next, integration by parts implies that total output is given by

$$\begin{aligned} Y^i(R^i) &= \int_{R^i}^{\frac{\alpha^i}{\beta^i}} x f^i(x) dx \\ &= \frac{(\alpha^i + \beta^i R^i)(\alpha^i - \beta^i R^i)}{2\beta^i}. \end{aligned} \tag{B.3}$$

Appendix C Numerical Implementation

The stationary equilibria of the two regimes and the non-stationary equilibrium of the transition phase are solved on a grid of 80 points in E with a time step of 0.01 years. The welfare comparisons of Section 6 are computed by Monte Carlo simulation of the equilibrium dynamics. For each transition length on the grid $T \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1, \dots, 8, 10\}$ years and each of ten initial capitalizations E_0 spanning $[\underline{E}_{old}, \overline{E}_{old}]$, we simulate 72,000 paths of aggregate capital over a horizon of 100 years at the same time step, 8,000 paths for each of nine distinct random-number seeds, and evaluate $PV(Y)$ and $PV(S)$ along each path using the equilibrium policies of the two regimes and of the transition. Within each seed,

the same Brownian and Poisson draws are used for every T and E_0 , so that differences in welfare across transition lengths are computed with common random numbers; their Monte Carlo standard errors are the across-seed standard deviations of the differences divided by three.

The welfare-maximizing transition length T^* is the continuous maximizer obtained by parabolic interpolation of the simulated welfare through the maximizing grid point and its two neighbors; at the ends of the grid, the grid value is reported. Welfare at T^* is read off a shape-preserving interpolation of the simulated values. Welfare under immediate implementation ($T = 0$) is computed by switching the requirement at the announcement date on the same draws. One year's status-quo output, the unit used in Section 6, is $PV(Y)$ in the status quo divided by the present value of a unit annual flow over the simulation horizon.

At the baseline $\theta = 0.50$, the Monte Carlo standard errors of the welfare differences reported in Section 6 are below 0.04×10^{-3} . In Figure 8, the across-seed standard deviation of T^* with announcement is at most 0.11 years at the baseline and at most 0.2 years for $\theta \geq 0.43$; it reaches about one year only where T^* jumps between two local maxima and close to the threshold below which the tightening does not raise welfare. In the benchmark without announcement, the optimum at $E_0 = \underline{E}_{old}$ and at $E_0 = \overline{E}_{old}$ is an immediate implementation in every seed; at $E_0 = 0.15$, the standard deviation of T^* is 0.12 years at the baseline, at most 0.2 years for $\theta \geq 0.47$, and rises to 1.8 years towards the benchmark's threshold.

Table 4 reports the optimal transition lengths and the welfare numbers of Section 6 for all ten initial capitalizations. Where the benchmark's optimum is an immediate implementation, the last three columns coincide by construction; where it is interior, welfare with the anticipation effect at the benchmark's transition length is read off the same interpolation.

Table 4: Welfare Gain, Timing, and Anticipation by Initial Capitalization

E_0	T^* (years)		Welfare gain (percent)	Share of the welfare gain lost (percent)		
	announced (1)	benchmark (2)		immediate (4)	no anticipation (5)	anticipation ignored (6)
0.062	3.1	0.0	2.13	12	12	12
0.085	3.3	0.0	2.11	39	39	39
0.108	3.3	2.2	2.11	68	48	4
0.130	3.3	1.7	2.12	86	47	9
0.153	3.3	1.3	2.15	72	46	16
0.176	3.1	0.8	2.18	56	43	26
0.199	2.9	0.4	2.22	41	39	35
0.221	2.4	0.0	2.28	27	27	27
0.244	2.1	0.0	2.41	16	16	16
0.267	1.6	0.0	2.54	6	6	6

Note. Baseline $\theta = 0.50$. The first and the last row are $E_0 = \underline{E}_{old}$ and $E_0 = \overline{E}_{old}$. Columns (1) and (2) give the welfare-maximizing transition length with announcement and in the benchmark without announcement. Column (3) is the welfare gain from the tightening under the optimal announced transition relative to the status quo, in percent of one year's output in the status quo. Columns (4) to (6) give the share of this gain that is lost under an immediate implementation, under the best transition without announcement (the contribution of the anticipation effect), and when the benchmark's transition length is announced and implemented (the cost of ignoring the anticipation effect).